

**International Institute of Information Technology
Hyderabad**

Computer Science and Engineering

EcoTwin

**Ecosystem Analog Search and Forecasting Using
Geospatial Foundation Models**

Project Report

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Abstract

EcoTwin is an end-to-end geospatial ecosystem analog discovery platform designed around the question: *which ecosystems elsewhere exhibit similar spatial structure, and what can their historical trajectories tell us about a target ecosystem?* HydroLAKES polygons provide ecosystem boundaries; a buffered, tiled Sentinel-2 acquisition pipeline through Google Earth Engine yields one-kilometre analysis cells, converted into six-band Prithvi embeddings, coverage-pooled to one 768-dimensional lake-year vector and indexed in PostgreSQL/pgvector. A query ecosystem is compared as a *five-year trajectory* rather than a single snapshot, sliding across each candidate’s history to find its best-matching window, and the ranked results are exposed through an interactive map and profile views with historical index trajectories.

The deployed system holds a complete, fully processed cohort — **829 active lakes across 26 Indian states**, ten consecutive years (2016–2025), **8,227 lake-year embeddings**, **19,046** imagery tiles, and **233,002** analysis cells, with no outstanding failures at any pipeline stage. The forecasting layer is analog-based and non-parametric: rather than decoding an embedding back into reflectance, it uses the embedding space only to *rank* analogs, then transfers their measured year-on-year changes onto the target as a rank-weighted directional forecast, traceable to the specific lakes, years, and deltas that produced it. The system is demonstrated on five Ramsar-designated wetlands and reservoirs — Chilika, Maharana Pratap Sagar (Beas/Pong Dam), Tawa Reservoir, Tso Morari, and Sasthamcotta — reported in full in Chapter 6.

Keywords: Geospatial Foundation Models, Prithvi, Remote Sensing, Sentinel-2, HydroLAKES, Ecosystem Analogs, Vector Search, pgvector, NDCI, Red-Edge Indices, Ramsar Wetlands, Trajectory Similarity

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Chapter 1

Introduction

1.1 Background and Problem Statement

Earth observation satellites provide repeated measurements of ecosystems at spatial scales that cannot be obtained economically through field observations alone, and Sentinel-2’s multispectral optical imagery is well suited to vegetation, water, land-cover, and surface-condition analysis. Rather than beginning with a specific question such as vegetation change or water extent, EcoTwin treats ecosystem comparison as a retrieval problem: satellite observations are converted into learned feature vectors and indexed, so a lake can be compared against a large reference collection without repeatedly processing raw imagery, asking whether another ecosystem looks similar in its spatial-spectral representation.

Several engineering constraints shape how this is actually solved. Lake boundaries are irregular polygons rather than fixed-size image squares, and large satellite scenes are expensive to download and store, so large ecosystems may require multiple image tiles rather than a single export. Historical Sentinel-2 collections do not expose identical quality bands across all years, and cloud or invalid-pixel contamination must be controlled for regardless of collection; every resulting embedding must remain associated with its source geometry and lake/year metadata rather than existing as an anonymous vector, and online similarity search must be substantially cheaper than recomputing embeddings from scratch.

1.2 Core EcoTwin Hypothesis

EcoTwin is based on the operational hypothesis:

Ecosystems elsewhere that are structurally similar in geospatial feature space can provide useful analogs for understanding the spatial and historical behaviour of a target ecosystem.

The hypothesis separates two tasks. The first is **analog discovery**: finding ecosystems that resemble the target in the learned spatial-spectral representation. The second is

trajectory analysis: comparing the historical environmental indicators of the target against those of its discovered analogs.

1.3 Project Evolution and Contributions

The implementation evolved from an initial design of standardized 5×5 km ecosystem blocks toward a lake-centred buffered imagery stage, local raster merging, and 1×1 km analysis cells: the larger imagery tiles are an I/O and Earth Engine management mechanism, while the smaller cells are the units from which embeddings are generated, which lets large lakes be processed without forcing the foundation model to operate directly on irregular polygons.

Two further pivots shaped the final system. First, **snapshot similarity gave way to trajectory similarity:** comparing single query and candidate embeddings proved too weak a notion of ecosystem resemblance, since two lakes can coincide in one year and diverge over a decade, so the implemented system instead compares a *five-year trajectory* of the query against every five-year sliding window in each candidate’s history, scoring by the best-matching window and retaining its position in time (Section 3.11). Second, an **embedding decoder was designed and then abandoned:** the original design intended to decode a predicted future embedding back into spectral indices, which would have required training an inversion of Prithvi’s 768-dimensional space — not calibrated, and exactly the kind of opacity the analog architecture exists to avoid. The implemented layer instead uses embeddings *only* to rank analogs, and computes predicted quantities from the analogs’ measured band statistics already in the database (Section 3.13): a transfer of observed change, not a reconstruction of an imagined image.

The project’s contribution is the resulting end-to-end lake-to-embedding-to-analog pipeline, executed to completion over an 829-lake, ten-year cohort with no residual failures: a satellite-image acquisition strategy built on Google Earth Engine and local raster processing with polygon-aware spatial subdivision and coverage filtering; Prithvi-based six-band representations with coverage-weighted pooling from cell to lake level, persisted as 768-dimensional vectors and retrieved through trajectory-based analog search; the decoder-free forecasting method above; a band-honest ecological index set computed only from channels the imagery actually contains; and a FastAPI/React-Leaflet application with automated PDF reporting, evaluated on five Ramsar-designated wetlands and reservoirs.

1.4 Related Work

EcoTwin sits at the intersection of three research threads that are each well-established individually but, to the authors’ knowledge, not previously combined into one lake-scale retrieval-and-forecast system: analog-based forecasting, geospatial foundation models used for embedding similarity, and remote-sensing indices with published in-situ validation. This section places EcoTwin against each in turn — first work that pursues a similar *method*, then work that offers independent grounds to trust

EcoTwin’s *measurements*.

1.4.1 Analog-Based and Case-Based Forecasting

The idea of forecasting a system’s future by finding its closest historical match and following what that match subsequently did is not new; it is the *analog method*, and EcoTwin’s twin-lake retrieval (Section 3.11) is a direct descendant of it. The method’s classical formulation is meteorological: Lorenz (1969) searched five years of Northern Hemisphere pressure fields for two states similar enough to serve as an analog pair, and found that even among a large historical record, truly close analogs were rare, with prediction error still growing at a rate comparable to the atmosphere’s intrinsic unpredictability.¹ Van den Dool (1989) extended this into operational weather forecasting through analogues, while separately estimating that a library on the order of 10^{30} years of observations would be needed to find true analogs matching to within observational error over a hemispheric domain — a quantitative statement of exactly the problem EcoTwin’s *trajectory* matching (five years of embeddings, not one snapshot) and *cross-lake* pooling (828 candidate histories, not one lake’s own past) are structural attempts to work around: a single lake’s own ten-year record could never contain a good analog of itself, so the search is deliberately widened to embedding-space neighbours drawn from an entire cohort.² Outside meteorology, analogy-based forecasting has also been formalised as a general forecasting technique — *structured analogies*, in which multiple historical cases are identified, matched, and combined by rule rather than judgement — which is methodologically close to EcoTwin’s fixed-weight, rank-ordered twin combination (Section 3.13), applied here to ecosystem reflectance trajectories rather than to the general business and policy forecasts the technique was originally developed for.³

1.4.2 Geospatial Foundation Models and Embedding-Based Ecosystem Retrieval

EcoTwin’s use of Prithvi (Section 3.9) sits within a rapidly growing body of work applying geospatial foundation models (GeoFMs) to Earth observation. Prithvi-EO-2.0 itself, released by a NASA/IBM-led consortium of 42 authors across 12 institutions, reports state-of-the-art downstream performance across benchmark tasks including flood mapping, burn-scar detection, and multi-temporal land-cover classification, outperforming six other GeoFMs it was benchmarked against.⁴ A separate, independent evaluation of the original Prithvi model’s domain adaptability found its embeddings competitive on high-level image-analysis benchmarks and proposed band-adaptation and multi-scale fine-tuning strategies to improve cross-domain transfer — external

¹E. N. Lorenz, “Atmospheric Predictability as Revealed by Naturally Occurring Analogues,” *Journal of the Atmospheric Sciences*, vol. 26, no. 4, pp. 636–646, 1969.

²H. M. van den Dool, “A New Look at Weather Forecasting through Analogues,” *Monthly Weather Review*, vol. 117, no. 10, pp. 2230–2247, 1989.

³K. C. Green and J. S. Armstrong, “Structured analogies for forecasting,” *International Journal of Forecasting*, vol. 23, no. 3, pp. 365–376, 2007.

⁴D. Szwarcman *et al.*, “Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation Applications,” arXiv:2412.02732, 2024.

evidence that Prithvi’s learned representation is sound enough to build on, from a group with no stake in EcoTwin’s own results.⁵

Closer to EcoTwin’s specific use of a GeoFM — similarity in *embedding space* as the basis for ecosystem-level analysis, rather than pixel classification — is a growing line of work built on Google DeepMind’s AlphaEarth embeddings. Ryan *et al.* (2026) compare an AlphaEarth-embedding-based wetland vegetation classifier against a conventional multi-sensor pipeline at Narran Lake, Australia, and report comparable accuracy (overall accuracy 0.985–0.991) with a simpler, cleaner workflow, while a related study applies the same embeddings to atoll island ecosystem mapping. Neither performs trajectory matching or forecasting — both are single-snapshot classification — but both independently validate the premise EcoTwin is built on: that a modern GeoFM’s embedding space encodes ecologically meaningful structure that similarity operations can exploit directly, without hand-engineered spectral rules, for wetland-type ecosystems specifically.⁶ EcoTwin differs from this line of work in two respects that follow directly from its forecasting goal rather than a classification goal: it compares five-year *trajectories*, not single images (Section 3.11), and it never asks the embedding space to output a class label or a reconstructed value, only a ranking (Section 3.10).

1.4.3 Independent Validation of the Ecological Indices

Section 3.13.1 and the per-site *Documented Condition* subsections of Chapter 6 ask whether EcoTwin’s specific, measured results agree with independently published information about the same lakes. A complementary question is whether the *index formulas* themselves — as opposed to any one lake’s trajectory — are trustworthy in the first place. NDCI, the index EcoTwin uses for chlorophyll-a/algal bloom (Table 2.3), is the most extensively validated of the five: a 2026 review synthesising validation evidence across 122 published studies of Sentinel-2 chlorophyll-a monitoring reports NDCI achieving good agreement with in-situ chlorophyll-a measurements (correlation coefficients in the 0.8+ range) in eutrophic inland and coastal waters, while also flagging a known limitation directly relevant to this report: NDCI’s relationship with chlorophyll-a becomes nonlinear and less reliable below roughly 12 $\mu\text{g/L}$, i.e. in clear, oligotrophic water — exactly the regime of a site like Tso Morari (Chapter 6), where NDCI readings should accordingly be read as a directional signal rather than a precise concentration proxy.⁷ NDVI and NDWI, the two indices most affected by this report’s switch from a B7 proxy to true-NIR B8 (Section 4.8), rest on the older and more extensively reviewed vegetation-index literature already cited in this report’s bibliography (Pettorelli *et al.*, 2005; Huete *et al.*, 2002) rather than a site-specific validation study, since both are standard, widely-adopted formulas rather than novel constructions specific to this project.

⁵C.-Y. Hsu, W. Li, and S. Wang, “Geospatial foundation models for image analysis: evaluating and enhancing NASA-IBM Prithvi’s domain adaptability,” arXiv:2409.00489, 2024.

⁶S. Ryan, M. Powell, J. Ling, and L. Wen, “Streamlining Wetland Vegetation Mapping with AlphaEarth Embeddings: Comparable Accuracy to Traditional Methods with Cleaner Maps and Minimal Preprocessing,” *Remote Sensing*, vol. 18, no. 2, article 293, 2026.

⁷S. Goodrich *et al.*, “Sentinel-2 for chlorophyll-a water quality monitoring: a review of validation evidence and application potential,” *International Journal of Remote Sensing*, 2026. <https://doi.org/10.1080/01431161.2026.2637851>.

None of the work surveyed here validates EcoTwin's *forecasts* directly — that would require the hindcasting protocol of Section 3.13.3, not yet run for the reasons given there. What it does provide is independent grounding for the two components the forecast is built from: that analog-based reasoning is a long-established, quantitatively understood forecasting paradigm rather than an ad hoc choice, and that both the foundation model supplying the similarity signal and the index formulas supplying the measured deltas have been separately evaluated, by researchers with no connection to this project, and found to carry real signal.

Chapter 2

Data Sources and Data Engineering

2.1 Overview

EcoTwin combines static geographic information (HydroLAKES polygons and meta-data) with repeated satellite observations (Sentinel-2), so that the static layer defines *where* an ecosystem is while the satellite layer defines *what it looks like through time*. The database representation retains the identifying, geographic, and descriptive fields it needs downstream — lake identifier and name, country, polygon geometry, area, elevation, pour-point coordinates, lake type, average depth, total volume, and watershed area — summarised with their pipeline role in Table 2.1. The complete HydroLAKES-derived inventory is much larger than the processing cohort: the pipeline was narrowed to a selected subset of lakes so that multi-year imagery and embeddings could be generated within available storage, Earth Engine, and compute budgets.

Table 2.1: HydroLAKES-derived information used by EcoTwin.

Field	Type	Role in EcoTwin
Lake ID	Integer	Primary ecosystem identifier
Lake name	Text	Human-readable identification
Country	Text	Geographic filtering
Lake area	Numeric	Size and metadata analysis
Elevation	Numeric	Environmental context
Pour latitude/longitude	Numeric	Geographic reference
Geometry	PostGIS geometry	Spatial processing and mapping
Lake type	Categorical	Ecosystem metadata
Average depth	Numeric	Descriptive context
Total volume	Numeric	Descriptive context
Watershed area	Numeric	Catchment context

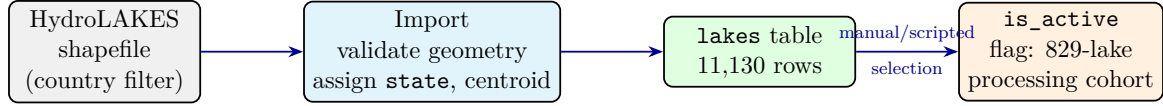


Figure 2.1: Stage 1 (Sync Lakes): the HydroLAKES shapefile is imported once per country into the full `lakes` inventory (11,130 rows), and `is_active` marks the 829-lake processing cohort that the remaining six stages operate on.

2.2 Sentinel-2 and Temporal Coverage

Sentinel-2 multispectral imagery, filtered, composited, and exported through Google Earth Engine, is the primary remote-sensing input. The principal collection is `COPERNICUS/S2_SR_HARMONIZED`, with a fallback to `COPERNICUS/S2` where the surface-reflectance harmonized collection provides no observations — particularly important for early Sentinel-2 years.

Ingestion was extended from an initially planned 2018–2025 window back to 2016, giving ten consecutive years. Table 2.2 shows the result: from 2018 onward every one of the 829 active lakes has a complete, embedded, feature-extracted record, while 2016 and 2017 fall slightly short because Sentinel-2B did not launch until March 2017, leaving a single-satellite, longer-revisit period in which some lakes lacked enough cloud-free acquisitions for a usable annual composite.

Table 2.2: Achieved temporal coverage across the 829-lake cohort. Every retained lake-year carries both an embedding and a completed feature record; there are no partial rows.

Year	Lake-years	Embeddings	Features
2025	829	829	829
2024	829	829	829
2023	829	829	829
2022	829	829	829
2021	829	829	829
2020	829	829	829
2019	829	829	829
2018	829	829	829
2017	808	808	808
2016	787	787	787
Total	8,227	8,227	8,227

The 42 absent lake-years are recorded as such rather than interpolated, since the similarity layer requires five consecutive years to form a comparison window: a lake missing 2016 simply contributes fewer candidate windows rather than a fabricated one.

The embedding input uses six bands, B2–B7 (blue, green, red, narrow near-infrared, and two short-wave infrared channels), selected consistently from the exported GeoTIFFs before model inference.

2.3 Environmental Indices: A Band-Honest Index Set

The merged lake imagery used for Prithvi inference contains only B2–B7 — not B8 (broad NIR), B11, or B12 (SWIR) — so the textbook forms of NDVI and NDWI, which both require B8, could not be computed from that raster alone. During initial development, before B8 statistics existed for the cohort, EcoTwin substituted the narrow-NIR red-edge channel B7 (783 nm) for the missing B8 and recorded the substitution in the index names (NDVI-B7, B7-based NDWI) rather than concealing it. That gap has since closed: the server-side B8 pipeline of Section 4.8 now populates `lake_features.b8_mean` for every lake-year in the cohort (8,227/8,227, Table 5.2) without ever downloading a raster, so NDVI and NDWI are computed with the standard formulas against true NIR. Only NDCI, Turbidity, and Red-Edge Slope — which never needed NIR — are unaffected by this change. All five indices are computed from the per-band *mean* reflectance of each lake-year over pixels inside the lake polygon: NDCI ($\frac{B5-B4}{B5+B4}$), the primary eutrophication/algal-bloom indicator, since B5 (705 nm, red edge) is where chlorophyll-a reflectance rises sharply while B4 (665 nm) remains absorbed; NDVI ($\frac{B8-B4}{B8+B4}$), the standard vegetation-vigor formula, tracking floating and emergent vegetation — macrophytes, water hyacinth, and the *phumdi* mats of some Indian wetlands — now against true NIR rather than the earlier red-edge proxy; NDWI ($\frac{B3-B8}{B3+B8}$), the standard McFeeters water index, which falls as vegetation raises B8 and so acts as NDVI’s natural mirror signal; Turbidity ($\frac{B4-B3}{B4+B3}$), which rises with suspended sediment; and Red-Edge Slope ($\frac{B7-B5}{78nm}$), the gradient across the red-edge region, sensitive to pigment concentration and canopy density independently of the normalized-difference family.

Table 2.3: Implemented ecological index set and ecological meaning.

Index	Formula	Ecological interpretation
NDCI	$(B5 - B4)/(B5 + B4)$	Chlorophyll-a; eutrophication and algal bloom intensity
NDVI	$(B8 - B4)/(B8 + B4)$	Vegetation vigor (standard formula); macrophyte and floating-weed encroachment
NDWI	$(B3 - B8)/(B3 + B8)$	Open-water extent (standard McFeeters formula); falls as vegetation colonises the surface
Turbidity	$(B4 - B3)/(B4 + B3)$	Suspended sediment load and water clarity
Red-edge slope	$(B7 - B5)/78 \text{ nm}$	Pigment concentration and canopy density trend

All five are computed in `app/services/index_calculator.py`, which returns `None` for any index whose input bands are missing rather than substituting a default — so a gap in the data propagates as a gap, never as a spurious zero.

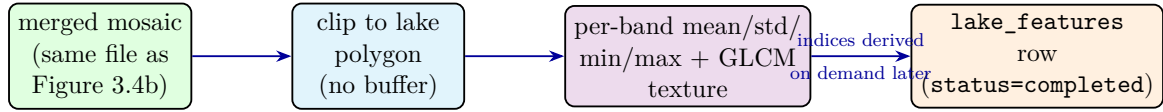


Figure 2.2: Stage 6 (Generate Lake Features): unlike Prithvi inference, this stage clips the merged mosaic to the bare lake polygon (no buffer, no cell grid) and computes per-band statistics and GLCM texture directly; only the band means are persisted, and each index of Table 2.3 is recomputed on demand (Section 3.13).

B8 is deliberately excluded from the Prithvi input, whose band count and normalization constants are fixed by the model’s training configuration, but is useful as an independent descriptor, so it is acquired separately (Section 4.8) and stored alongside the other band statistics for every lake-year.

A valid ecosystem record requires a valid lake polygon and year, a successful imagery query and usable composite, valid exported raster data, valid spatial cells after coverage filtering, and a successfully generated embedding for each retained cell; the pipeline distinguishes successful, failed, skipped, and already-completed work so that long-running jobs can be resumed.

Chapter 3

Detailed EcoTwin Methodology

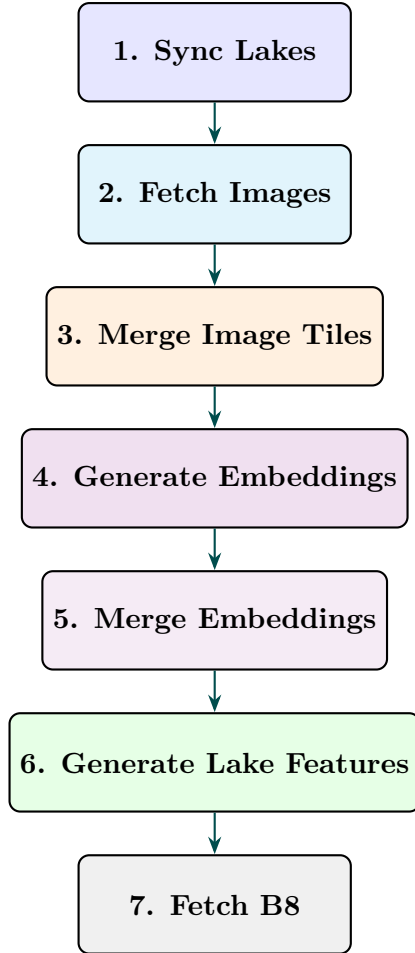
3.1 End-to-End Pipeline

The implemented workflow is:

$$\begin{aligned} & \textit{Lake Inventory} \rightarrow \textit{Boundary Validation} \\ & \quad \rightarrow \textit{Buffering} \rightarrow \textit{Imagery Tiling} \\ & \quad \rightarrow \textit{Sentinel-2 Composite} \rightarrow \textit{GeoTIFF} \\ & \quad \rightarrow \textit{Local Mosaic} \rightarrow \textit{1 km Cells} \\ & \quad \rightarrow \textit{Coverage Filter} \rightarrow \textit{Prithvi} \\ & \quad \rightarrow \textit{768D Embeddings} \rightarrow \textit{pgvector} \\ & \quad \rightarrow \textit{Cosine Search} \rightarrow \textit{Analog} \\ & \quad \rightarrow \textit{Temporal Profiles} \rightarrow \textit{Forecast} \end{aligned} \tag{3.1}$$

3.2 Data-Fetching Workflow: The Seven Operator-Triggered Stages

The seventeen-step scientific pipeline above is what EcoTwin *computes*; Figure 3.1 shows how an operator actually *runs* it, as the seven sequential stages exposed by the control strip of Figure 4.1. Each stage is a separate API call, a separate FastAPI `BackgroundTasks` job, and a separate database write — so a failure at stage 4 leaves stages 1–3 fully intact and does not require re-fetching imagery already on disk.



- 1. Sync Lakes** — import HydroLAKES into the `lakes` table (Figure 2.1).
- 2. Fetch Images** — buffer, tile, and export Sentinel-2 GeoTIFFs per lake-year (Steps 1–8 below).
- 3. Merge Image Tiles** — mosaic a lake-year’s tiles into one raster (Step 9).
- 4. Generate Embeddings** — split the mosaic into cells and run Prithvi inference (Steps 10–12).
- 5. Merge Embeddings** — pool cell embeddings into one lake-year vector (Step 12a).
- 6. Generate Lake Features** — compute per-band statistics and ecological indices (Step 16, Figure 2.2).
- 7. Fetch B8** — server-side `reduceRegions()` enriches `lake_features` with the auxiliary NIR band (Figure 4.3).

The remainder of this chapter walks through these seven stages at step-by-step granularity, numbered independently of the stage list above.

Figure 3.1: The seven-stage data-fetching workflow, in the order an operator triggers them via the control strip (Figure 4.1). Each stage reads the previous stage’s database output as its own input, never the raw upstream source directly — stage 3 reads `lake_tiles`, not Earth Engine again; stage 6 reads the merged mosaic already on disk, not a fresh GEE export. Stage 7 (B8) is independent of stages 4–5 and can run in parallel with them, since it enriches existing `lake_features` rows rather than depending on embeddings.

Every stage is **idempotent at the row level**: each writes a status column (`pending/processing/completed/ failed`) before doing any work, and a re-run skips rows already `completed` rather than reprocessing them. Stage 7 (B8) has no code dependency on stages 4–5 despite the drawn order, since B8 statistics are written directly onto an existing `lake_features` row, so it can be fetched before, after, or concurrently with embedding generation.

3.3 Steps 1–2: Lake Boundary Retrieval and Buffering

The pipeline reads the target lake polygon from PostGIS, keeping the original geometry in a spatial database to avoid repeated conversion between shapefiles and application-

specific formats, checks it for validity, and transforms it into a projected CRS. The polygon is then expanded using a configurable buffer (1200 m during development), since the ecosystem representation is not restricted to the water polygon itself. The surrounding land, shoreline, vegetation, and local landscape context contribute to the spatial-spectral representation.

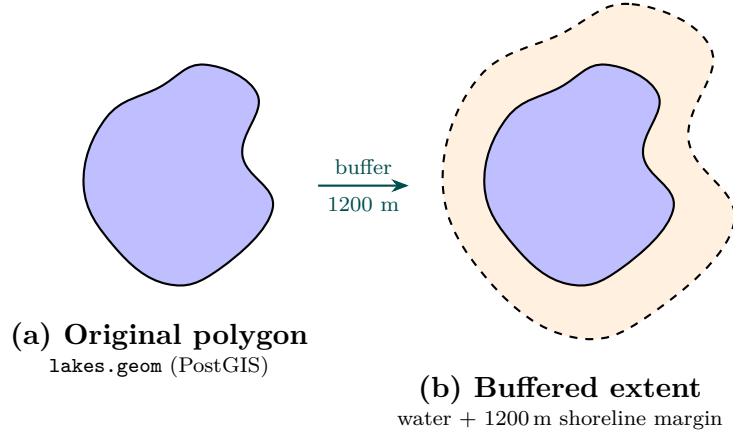


Figure 3.2: Step 2 (Buffering): the original polygon (a) is expanded by a fixed 1200 m buffer (b); everything between the two boundaries is retained as part of the ecosystem representation, not discarded as “outside the lake.”

3.4 Step 3: Projected Grid Generation

The buffered polygon is projected into a metric CRS. A regular imagery grid is then generated around the buffered geometry.

The imagery grid is deliberately different from the final analysis grid. During the latest extraction implementation, an approximately 8500 m tile size was used to balance Earth Engine request size, memory use, and local processing cost.

A lake may therefore produce one, two, or several imagery tiles depending on its size and shape.

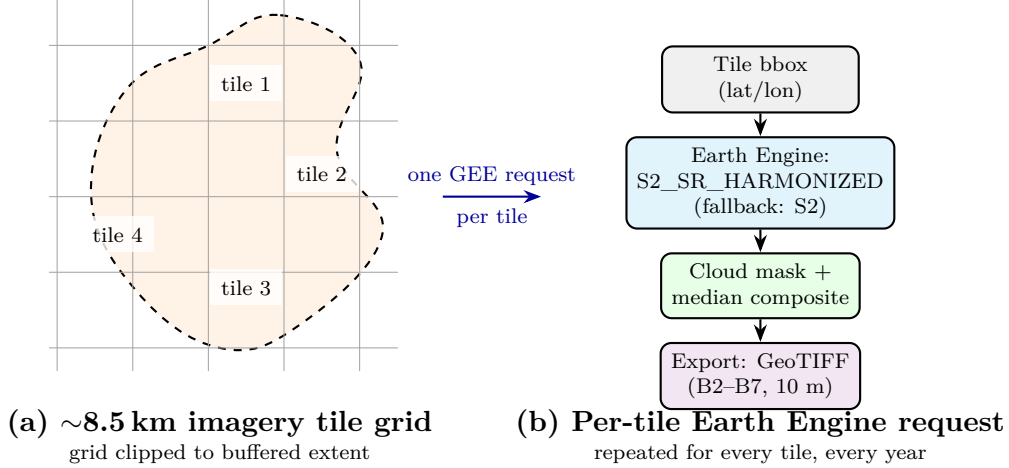


Figure 3.3: Step 3 (Imagery Tiling): the buffered extent (a) is clipped against a ~8.5 km grid (here four tiles); each tile drives one Earth Engine request (b) — the “Fetch Images” data-fetching stage of Figure 3.1.

3.5 Steps 4–8: Sentinel-2 Compositing, Export, and Resumability

For each lake-year combination, the pipeline first queries the preferred surface-reflectance collection; if no usable observation exists, it falls back to the Level-1C collection (Table 3.1). This decision is made before the export request and is recorded in the processing metadata.

Table 3.1: Sentinel-2 collection strategy.

Priority	Collection	Purpose
1	COPERNICUS/S2_SR_HARMONIZED	Preferred surface reflectance
2	COPERNICUS/S2	Fallback for early or unavailable surface-reflectance years

Scenes are filtered before compositing: the deployed configuration excludes those whose reported cloud cover exceeds 20% (`GEE_CLOUD_COVER_MAX`) and masks individual unusable pixels using cloud-probability and class-based information where available — collection-aware, since early Sentinel-2 products expose different quality bands (`QA60`, `MSK_CLASS1_OPAQUE`, `MSK_CLASS1_CIRRUS`) than later ones. The filtered observations are then combined into an annual per-pixel, per-band median composite,

$$I_{\text{annual}}(p, b) = \text{median}(I_1(p, b), I_2(p, b), \dots, I_n(p, b)) \quad (3.2)$$

clipped to the requested extent, and exported as local GeoTIFF at 10 m resolution in floating-point form, so the same composite never needs recomputing for a later analysis cell. Each tile carries status metadata (lake ID, year, tile index, bounding box,

local path), and the pipeline checks this before starting a new Earth Engine request for the same tile — concrete, tile-level idempotency (Section 3.2): a batch interrupted partway through a lake resumes from its first incomplete tile.

3.6 Step 9: Local Raster Merging

After all tiles belonging to a lake-year are available, rasterio merges them into a single raster mosaic — reading, reprojecting, and writing with compression settings that reduce storage while preserving the numerical data model inference needs. The merged image becomes the stable source from which smaller analysis cells are extracted.

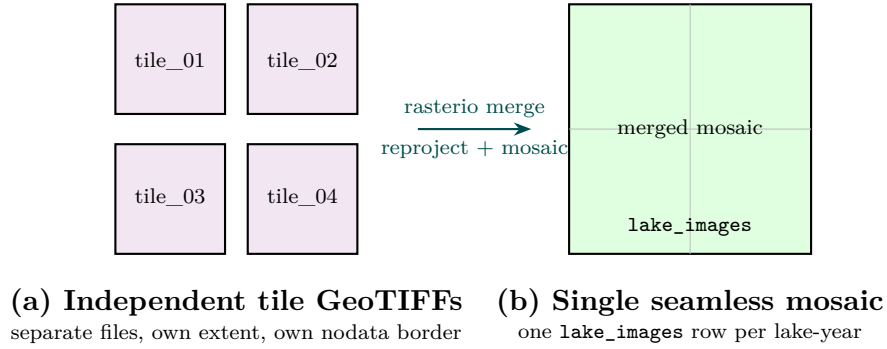


Figure 3.4: Step 9 (Local Raster Merging): independent tile GeoTIFFs (a) are mosaicked by `rasterio.merge` into one seamless lake-year raster (b), recorded in `lake_images` — the “Merge Image Tiles” stage of Figure 3.1.

3.7 Step 10: Analysis-Cell Generation

The merged lake image is intersected with a regular 1×1 km analysis grid, and for each cell the lake coverage is computed,

$$Coverage(c) = \frac{Area(c \cap Lake)}{Area(c)} \times 100 \quad (3.3)$$

with only cells above a 15% threshold retained, which prevents cells that only touch a small corner of a lake from being treated as representative ecosystem regions.

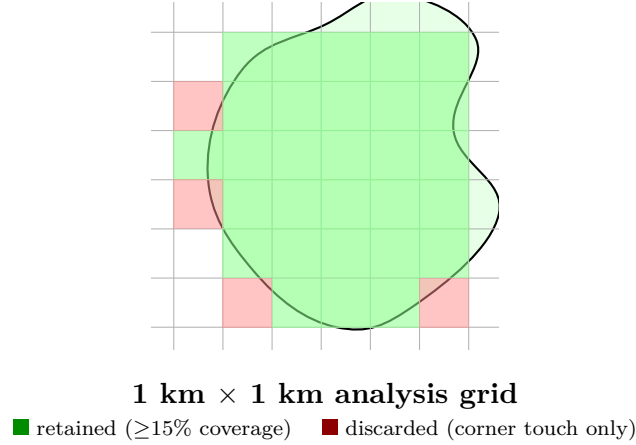


Figure 3.5: Step 10 (Analysis-Cell Generation): the merged mosaic is intersected with a 1 km grid; cells with substantial coverage (green) are retained as embedding units, corner-touch cells below 15% (red) are discarded.

3.8 Step 11: Model Input Preparation

Each retained analysis cell is read from the local merged GeoTIFF and the six selected bands are converted into the model’s expected tensor layout, numerical range, and spatial dimensions. Separating geographic geometry, raster extraction, model preprocessing, and model inference into distinct stages makes it possible to change the model without re-downloading the underlying satellite data.

3.9 Step 12: Prithvi Embedding Generation

Prithvi is used as the geospatial foundation model: each retained analysis cell, as a six-band B2-B7 image X , is transformed into a fixed-length feature vector $f(X) \rightarrow \mathbf{z} \in \mathbb{R}^{768}$, batched eight at a time (`EMBEDDING_GPU_BATCH_SIZE`) on CPU inference. Every output vector is L2-normalised, so that for unit-norm vectors the cosine similarity reduces exactly to the inner product and the entire similarity search can be executed as matrix multiplication.

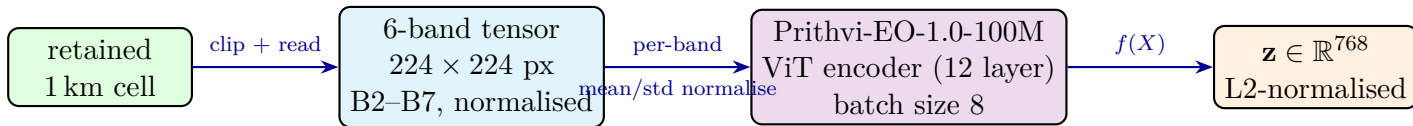


Figure 3.6: Steps 11–12: each retained cell is clipped and read as a 224×224 -pixel, six-band tensor, normalised using Prithvi’s own per-band mean/std (`PRITHVI_BAND_MEAN/_STD`), then batched into the Prithvi-EO-1.0-100M encoder — the only stage where the foundation model is invoked — producing a 768-d, L2-normalised vector.

3.10 Step 12a: Coverage-Weighted Lake-Level Pooling

Cell-level embeddings preserve spatial detail, but retrieval and forecasting operate on lakes, so for a lake-year with cells $c_1, \dots, c_N$ having embeddings $\mathbf{e}_i$ and lake-coverage percentages w_i , the lake embedding is the coverage-weighted mean, re-normalised:

$$\mathbf{z}_{lake} = \frac{\sum_{i=1}^N w_i \mathbf{e}_i}{\left\| \sum_{i=1}^N w_i \mathbf{e}_i \right\|_2} \quad (3.4)$$

so that a cell almost entirely filled by the lake contributes more than one that merely clips its margin; cells with zero coverage, zero vectors, or the wrong dimensionality are excluded before pooling. Across the cohort this reduces **233,002** cell embeddings to **8,227** lake-year vectors, an average of **28.3** cells per lake-year, and the cell-level vectors are retained as the underlying evidence for any result the pooled vector produces.

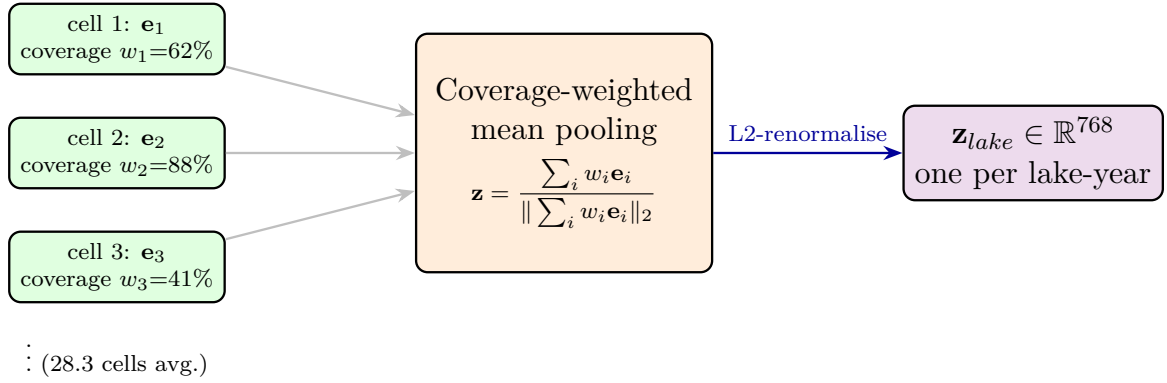


Figure 3.7: Step 12a (Coverage-Weighted Pooling) — the “Merge Embeddings” stage of Figure 3.1: each cell embedding contributes in proportion to its coverage w_i (e.g. 88% outweighs 41%), and the weighted sum is L2-renormalised for comparability in regions.

Before insertion into the database, every embedding is checked for expected dimensionality, missing or null values, numerical validity, and correct lake, year, and spatial-cell association; a failed inference is not inserted as a valid vector. The final embedding is then stored in PostgreSQL using pgvector under the conceptual relationship Lake $\rightarrow$ Year $\rightarrow$ Spatial Cell $\rightarrow$ Embedding, which preserves the provenance required to explain why an ecosystem was retrieved as an analog.

3.11 Step 15: Trajectory Similarity Search

Comparing one query embedding against one candidate embedding asks only *which lake looked like this lake, in this year?* — too weak a question for forecasting, since two lakes can coincide in a single year and diverge over the following decade. EcoTwin instead asks which lake has been **moving through** the representation space the way

this lake has been moving: a shared five-year path is a far better basis for projecting the next three.

For a query lake with embeddings ordered by year, EcoTwin forms the trajectory ending at anchor year t_0 with window size $W = 5$,

$$Q = (\mathbf{q}_{t_0-4}, \mathbf{q}_{t_0-3}, \mathbf{q}_{t_0-2}, \mathbf{q}_{t_0-1}, \mathbf{q}_{t_0}) \quad (3.5)$$

and each candidate lake with at least W years of history is expanded into every contiguous five-year window (ten years of data yields six windows),

$$C^{(j)} = (\mathbf{c}_{y_j}, \dots, \mathbf{c}_{y_j+4}), \quad j = 1, \dots, n - W + 1 \quad (3.6)$$

which need not align in calendar time with the query: a lake that underwent in 2017–2021 the transition another lake is undergoing in 2021–2025 is precisely the analog sought, since it has already lived through the target’s near future. Each window is scored as the mean of the year-by-year cosine similarities between corresponding positions,

$$S(Q, C^{(j)}) = \frac{1}{W} \sum_{i=1}^W \frac{\mathbf{q}_i^\top \mathbf{c}_i^{(j)}}{\|\mathbf{q}_i\|_2 \|\mathbf{c}_i^{(j)}\|_2} \quad (3.7)$$

which (all embeddings being unit-norm) reduces to a mean of inner products, and each candidate lake is represented by its single best window,

$$S_{lake}^* = \max_j S(Q, C^{(j)}), \quad j^* = \arg \max_j S(Q, C^{(j)}) \quad (3.8)$$

whose end year $y_{j^*} + 4$ is the pivot on which the forecasting layer turns: the analog’s observed behaviour *after* that year is the evidence used to project the target’s future.

Because every vector is unit-norm and drawn from one model, raw cosine scores cluster tightly just below 1.0 (measured top analogs span roughly 0.986–0.999 in Chapter 6), which would make every analog read as “99% similar” if displayed directly. EcoTwin therefore linearly rescales $[\phi, 1.0] \rightarrow [0, 1]$ with floor $\phi = 0.5$,

$$S_{display} = \text{clip}_{[0,1]} \left(\frac{S^* - \phi}{1 - \phi} \right) \quad (3.9)$$

a monotone transform that changes no ranking; raw scores are reported alongside the rescaled ones in Chapter 6 so the distinction stays visible. Two further metrics are implemented and selectable per query (Euclidean distance mapped by $1/(1+d)$, and inner-product mapped by $(\langle \mathbf{a}, \mathbf{b} \rangle + 1)/2$), though cosine is the default used throughout this report.

The current implementation compares in Python across all candidate windows: with L lakes, n years, and dimension d , the cost is $O(L \cdot (n-W+1) \cdot W \cdot d)$, roughly $829 \times 6 \times 5 \times 768 \approx 1.9 \times 10^7$ multiply-accumulates per query — measured at **141–172 ms** in Chapter 6, comfortably interactive but linear in lake count, which motivates the pgvector ANN migration under future work.

3.12 Step 16: Temporal Profile Generation

For the query lake and its retrieved analogs, the stored band statistics are assembled into per-year index series using the formulas of Table 2.3, computed on demand from the persisted band means (the single source of truth for measured reflectance) rather than stored pre-computed, so a change to an index definition takes effect everywhere at once without a database migration.

3.13 Step 17: Directional Ecological Forecasting

The forecasting layer transfers *measured change* from the analogs onto the target, presented here in implementation order. The most recent completed feature row for the target lake fixes the anchor year t_0 and anchor values $v_k(t_0)$ for each index k ; forecasts are produced for $t_0 + 1, t_0 + 2, t_0 + 3$ (`_FORECAST_HORIZON`). The trajectory search of Section 3.11 then runs with an enlarged candidate pool, walked in rank order: a candidate is admitted as a *twin* only if it is a different lake than the target, has a completed feature row at its matched window end year, and has at least one usable year after that, so the five twins that emerge are the five best *usable* analogs rather than merely the five highest-scoring rows.

For twin i with matched window ending at year m_i , the change transferred at offset h is $\delta_{i,k,h} = v_{i,k}(m_i + h) - v_{i,k}(m_i)$ — not the analog’s absolute index value, which reflects its own climate, geology, and catchment, but the *change* it underwent, comparable across lakes in a way absolute values are not. Twins receive fixed weights by rank rather than weights derived from their similarity scores,

$$\omega_1 = 0.28, \quad \omega_2 = 0.24, \quad \omega_3 = 0.20, \quad \omega_4 = 0.16, \quad \omega_5 = 0.12 \quad (3.10)$$

precisely because raw similarity scores compress into a narrow band near 1.0 (Section 3.11), where score-proportional weights would be nearly uniform and discard the ranking the search worked to establish. Twins do not all have data at every offset, so with $\mathcal{A}_{k,h}$ the set of twins with a usable delta for index k at offset h , weights are renormalised over exactly that set,

$$\tilde{\omega}_{i,k,h} = \frac{\omega_i}{\sum_{j \in \mathcal{A}_{k,h}} \omega_j}, \quad i \in \mathcal{A}_{k,h} \quad (3.11)$$

so the denominator is always 1.0 and a year with only two twins does not silently shrink toward “stable”; the twin count is reported separately as confidence. The score for index k at offset h is

$$\Sigma_{k,h} = \sum_{i \in \mathcal{A}_{k,h}} \tilde{\omega}_{i,k,h} \delta_{i,k,h} \quad (3.12)$$

classified against threshold $\tau = 0.005$:

$$D_{k,h} = \begin{cases} \text{uncertain (?)} & |\mathcal{A}_{k,h}| < 2 \\ \text{up } (\uparrow) & \Sigma_{k,h} > +\tau \\ \text{down } (\downarrow) & \Sigma_{k,h} < -\tau \\ \text{stable } (\rightarrow) & \text{otherwise} \end{cases} \quad (3.13)$$

with *uncertain* checked first and kept distinct from *stable*: *stable* is a positive finding that the twins agree little changes, while *uncertain* admits too few twins had data to support any claim, and Chapter 6 contains real cases where this distinction carries the entire meaning of a result. An indicative magnitude $\hat{v}_k(t_0 + h) = v_k(t_0) + \Sigma_{k,h}$ accompanies each direction, reported as an expected value rather than a calibrated point prediction. Every forecast returns with a complete audit report — anchor values, weight scheme, classification rule, and each twin’s identity, matched window, and weighted contribution — so any published direction can be recomputed by hand from the reported numbers.

Unlike a parametric time-series model, whose prediction runs through learned parameters difficult to connect to a particular ecosystem, EcoTwin provides an explicit, inspectable chain: Target $\rightarrow$ Retrieved Twin $\rightarrow$ Similarity $\rightarrow$ Historical Trajectory $\rightarrow$ Projected Trajectory. There is no learned parameter anywhere in the method above; every quantity is either a measured reflectance-derived index or a fixed, disclosed constant ($\omega_{1.5}$ and τ), and an interval-based uncertainty measure (the spread across twin predictions) was considered and rejected in favour of the simpler twin count $|\mathcal{A}_{k,h}|$, since a naive dispersion measure over a small, unevenly-weighted set — as when Beas’s 2027 and 2028 red-edge scores each rest on a single twin’s trajectory (Chapter 6) — is not a stable confidence signal, while the count drops to zero exactly when a forecast should not be trusted and degrades gracefully otherwise.

3.13.1 Literature Survey: Why Not an LSTM-Based Forecast?

A recurrent-neural-network forecast of vegetation indices was considered and rejected, on the evidence of a directly relevant prior study: Reddy and Prasad (2018) train a Long Short-Term Memory (LSTM) network to predict MODIS NDVI one week ahead for two regions of Great Nicobar Island, using 861 weekly NDVI composites (2000–2016) split 70/20/10 into training, validation, and test sets, and report a test RMSE below 0.03 for both a seasonally-stable interior forest and a tsunami-disturbed coastal site.¹ This is a strong result, and it comes from the same lab that supervised this project — but three of its operating conditions are exactly the conditions EcoTwin does not have.

First, **data volume**: the cited LSTM is trained on 861 timesteps for a single site. EcoTwin has **ten** annual observations per lake (Table 2.2) — roughly two orders of magnitude less sequence per site than the training regime that produced the paper’s

¹D. Sushma Reddy and P. Rama Chandra Prasad, “Prediction of vegetation dynamics using NDVI time series data and LSTM,” *Modeling Earth Systems and Environment*, 2018, <https://doi.org/10.1007/s40808-018-0431-3>.

reported accuracy. An LSTM’s gates each carry their own weight matrices; fitting them from ten points per lake would fit noise, not seasonality. Second, **per-site versus cross-cohort**: the paper trains one network per region, retrained for each new site studied. EcoTwin runs against **829** lakes and cannot retrain a separate deep network for each one on ten data points; the entire premise of the analog approach (Section 3.11) is to substitute *other lakes’ history* for the sequence length a single lake does not have. Third, **forecast horizon**: the cited network predicts one step ahead (the next 7-day composite); its own conclusion explicitly separates this from a harder, unaddressed problem — forecasting “up to next 50 intervals” — and does not attempt it. EcoTwin’s requirement is a **1–3 year ahead** forecast from annual data, which is that harder, multi-step regime, not the one-step regime the cited result was measured on; naively chaining a one-step LSTM forward three times would compound prediction error at exactly the horizon EcoTwin needs to be honest about (Section 3.13.3).

A fourth reason is architectural rather than data-driven, and applies regardless of sequence length: an LSTM’s output is a function of learned weight matrices with no direct correspondence to a specific ecosystem, which is the same objection already raised against decoding a predicted embedding back into reflectance (Section 3.10) — both approaches route the forecast through parameters that cannot be pointed back at a particular lake, year, or measurement. The rank-weighted analog transfer used instead has no learned parameter at any stage (Section 3.13 above), so every forecast is traceable to specific twin lakes and measured deltas rather than to trained weights. None of this is a claim that LSTMs are unsuitable for vegetation time series in general — Reddy and Prasad’s results argue the opposite, for the regime they tested — only that the regime this project operates in (many lakes, few years each, multi-year horizon, and forecasts that must be traceable to their evidence) is a poor match for it.

3.13.2 Worked Example: Chilika, NDVI, $h = 1$

Chapter 6 reported Chilika’s 2026 NDVI forecast as $\uparrow$ with score $+0.0253$ from five twins. Reconstructing it: all five twins have a usable year at $h = 1$, so no renormalization is needed beyond the base weights (0.28, 0.24, 0.20, 0.16, 0.12), which already sum to 1. Each twin’s delta is its measured NDVI one year after its own matched-window end, minus its value at that end year — five independent measurements from five different lakes and calendar years, combined only through their rank weight. The result, $+0.0253$, exceeds $\tau = 0.005$, giving $\uparrow$, and every intermediate quantity is recoverable from the audit report returned alongside the forecast.

3.13.3 Hindcasting Protocol

A rigorous accuracy evaluation requires hindcasting: hide each lake’s final h_{max} years, run the identical retrieval and forecast pipeline using only years before the cutoff, and score the forecast against the held-out observations. For cutoff year $t_c \in \{2019, \dots, 2022\}$, both the query trajectory and every candidate’s embedding history are restricted to years $\leq t_c$; the method then runs exactly as deployed, producing $\hat{v}_k(t_c + h)$ for $h = 1, 2, 3$, compared against the actual recorded $v_k(t_c + h)$ already present in `lake_features`. The protocol reports, per index and per horizon (never

pooled, since scales differ by an order of magnitude or more), both a magnitude metric and a *directional hit rate*:

$$MAE_{k,h} = \frac{1}{n} \sum_{i=1}^n |v_{k,i}(t_c + h) - \hat{v}_{k,i}(t_c + h)| \quad (3.14)$$

$$HitRate_{k,h} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\text{sign}(\Sigma) = \text{sign}(\Delta_{actual})] \quad (3.15)$$

Running this at $t_c = 2022$ requires three held-out years (2023–2025) plus five years of prior history — exactly the ten years this project has, so a single cutoff gives only one evaluation point per lake: too coarse a sample for a defensible aggregate accuracy figure. Multiple cutoff years are not yet available because the dataset spans only one decade, so this protocol is documented here as the immediate next step (Section 8.2) rather than executed prematurely on insufficient data.

Chapter 4

System Architecture and Implementation

4.1 Architecture Overview

EcoTwin is implemented as a multi-component system. The offline pipeline performs the computationally expensive work, while the online application retrieves precomputed data.

Figure 4.1 shows the pipeline control strip from the deployed application. Each button triggers one of the seven offline stages of Figure 3.1 (Section 3.2), in the same execution order. Exposing each stage as an individually triggerable operation, rather than one opaque “run everything” action, is what makes the pipeline resumable (Step 8, Chapter 3): a failure in one stage does not require re-running the stages before it.

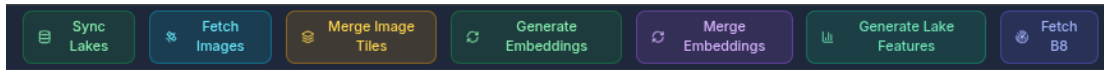


Figure 4.1: The deployed pipeline control strip. Each stage (lake sync, imagery fetch, tile merge, embedding generation, embedding merge, feature generation, B8 fetch) is independently triggerable and independently resumable.

4.2 Technology Stack

Table 4.1: EcoTwin implementation stack.

Layer	Technology	Responsibility
Frontend	React 18, Vite, TypeScript	Interactive application
Mapping	Leaflet / React-Leaflet	Spatial visualization
Charts	Recharts	Index trajectory plots
Client state	Zustand, TanStack Query	Local state and server cache
Backend	FastAPI, Python 3.12	REST services and orchestration
ORM / migrations	SQLAlchemy 2 (async), Alembic	Schema and access
Database	PostgreSQL 16	Relational persistence
Spatial database	PostGIS 3.4	Polygon and spatial queries
Vector search	pgvector	Embedding storage and similarity search
Remote sensing	Google Earth Engine	Satellite filtering/compositing
Raster processing	Rasterio, NumPy, Shapely	Local GeoTIFF processing
ML inference	Prithvi-EO-1.0-100M, PyTorch	Geospatial embeddings
Background work	FastAPI BackgroundTasks	Long-running pipelines
Logging	structlog	Structured pipeline logs
Reporting	Jinja2 + WeasyPrint	PDF report generation
Administration	pgAdmin	Database administration
Containerization	Docker Compose	Reproducible local deployment

4.3 Component Communication Workflow

Figure 4.2 traces how the five principal components — the React/Leaflet frontend, the FastAPI backend, the PostgreSQL/PostGIS/pgvector database, Google Earth Engine, and the Prithvi inference process — actually exchange data, distinguishing the calls that only ever happen offline (dashed, orange) from the calls a live user request makes (solid, blue).

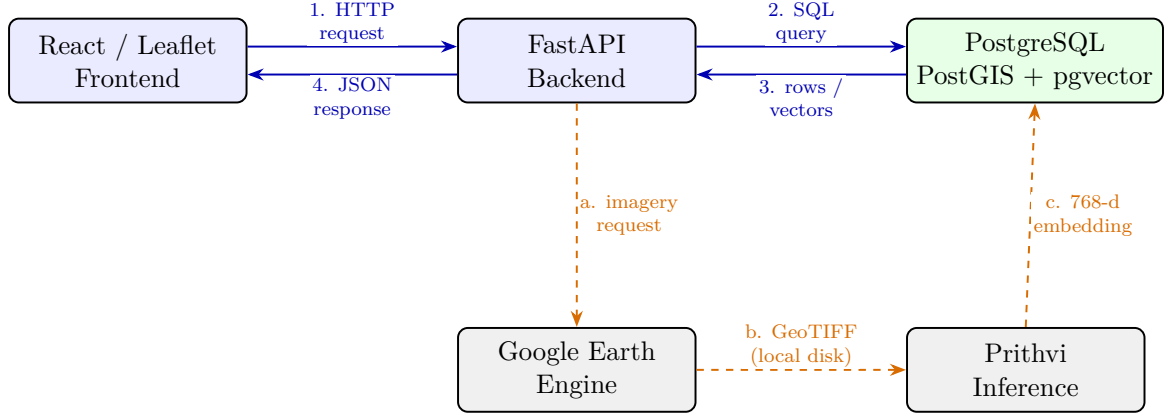


Figure 4.2: Component communication: solid blue is the online request path (frontend $\rightarrow$ backend $\rightarrow$ database, serving every request from precomputed rows); dashed orange is the offline pipeline path (backend $\rightarrow$ Earth Engine $\rightarrow$ Prithvi $\rightarrow$ database), triggered by Figure 4.1 as **BackgroundTasks**.

Neither path bypasses the backend: the frontend has no direct connection to the database, Earth Engine, or Prithvi, and the offline pipeline has no direct connection to the frontend. Every cross-component call passes through FastAPI, enforcing the offline/online separation in one place rather than by convention across every caller: the offline path (lake preparation, imagery acquisition, raster merging, spatial-cell extraction, embedding generation, temporal-feature preparation, vector indexing — the seven stages of Figure 3.1) is kept apart from the online path (lake lookup, map rendering, similarity query, analog profile retrieval, temporal-series retrieval, report generation), so Earth Engine and the foundation model never run during a user request.

4.4 PostgreSQL, PostGIS, and pgvector

PostgreSQL is the central persistent store. PostGIS keeps the original lake geometry, analysis-cell geometry, and spatial metadata in the database itself rather than converted into plain latitude/longitude tables, so intersection, containment, area, and map-query operations run directly as SQL. The pgvector extension stores the 768-dimensional Prithvi vectors alongside this spatial data, maintaining the association $\text{Embedding} \leftrightarrow \text{Subregion} \leftrightarrow \text{Lake} \leftrightarrow \text{Year}$ so that cosine-distance/similarity operations execute directly against the indexed database rather than exporting vectors to CSV or recomputing features.

4.5 Core Database Entities

The deployed schema, reached after nineteen Alembic migrations, consists of the tables in Table 4.2.

Table 4.2: Deployed database tables and current row counts.

Table	Rows	Role
<code>lakes</code>	11,130	HydroLAKES inventory with PostGIS geometry; <code>is_active</code> marks the 829-lake processing cohort
<code>lake_tiles</code>	19,046	Earth Engine export tiles per lake-year
<code>lake_images</code>	8,227	Merged lake-year mosaics with path and status
<code>sub_regions</code>	233,002	1 km analysis cells with geometry and coverage
<code>regions</code>	8,227	Pooled lake-year <code>vector(768)</code> embeddings
<code>lake_features</code>	8,227	Band statistics, indices, and texture per lake-year
<code>india_states</code>	36	State boundaries for spatial attribution
<code>reports</code>	—	Generated PDF report records

Despite its historical name (a legacy of the original design, preserved since renaming would invalidate existing embeddings for no functional gain), `regions` stores exactly one pooled 768-dimensional embedding per lake-year, enforced by a unique constraint on `(lake_id, year)` — the `vector(768)` column similarity search operates on. `lake_features`, the widest table in the schema, is the source of truth for measured reflectance: per-band statistics for B2–B7 and B8, pixel and coverage counts, cloud percentage, and GLCM texture, from which the forecasting layer reads only the band means and derives indices on demand. Keeping embeddings and measured reflectance in separate tables is what makes the decoder-free forecast possible: the embedding space answers *which lakes are analogs*, and the feature table independently answers *what measurably happened to them*, so the forecast never depends on inverting the foundation model.

4.6 Backend and Frontend

The FastAPI backend provides services for lake metadata lookup, nearest lake/grid-cell lookup, analog similarity search, temporal feature retrieval, forecast retrieval, report generation, and health/system status, separating cheap data retrieval from heavy batch processing so long operations are handled asynchronously.

The frontend is implemented using React, Vite, and TypeScript, with the map on Leaflet/React-Leaflet. A user selects or searches for a lake, views its geographic boundary and metadata, requests analogous ecosystems, visualizes their locations and similarity scores, and inspects their historical profiles alongside the forecast. Displayed lake boundaries and analog markers are drawn from the same PostGIS geometry used for the analysis itself, rather than a separately derived copy, which avoids the common failure mode where the analytical result and the displayed map disagree due to different coordinate transformations or numbering systems.

4.7 Asynchronous Processing and Progress Tracking

Imagery download, raster merging, feature extraction, and model inference all take far longer than an acceptable HTTP request, so each is launched through FastAPI’s `BackgroundTasks` and returns immediately. A distributed task queue was considered and not adopted: the pipelines are operator-run batch jobs on a single machine, and the durable state a queue would provide already exists in the database, where every tile, image, cell, and feature row carries its own status. Progress is instead exposed through module-level state guarded by a `threading.Lock` (processed, skipped, and failed counts, elapsed time, throughput, projected completion), with each pipeline’s block independent so concurrent runs report separately; the frontend polls the corresponding `/status` endpoint. Long jobs are also cancellable: a flag checked between work units means a cancelled run stops after the current lake-year rather than mid-write, and completed work stays committed.

4.8 Server-Side B8 Acquisition

Unlike B2–B7, which must be downloaded as pixels for Prithvi inference, B8 is needed only as summary statistics, so moving pixels at all would be wasted work. The B8 pipeline instead performs the reduction entirely on Earth Engine’s servers using `Image.reduceRegions()`: lake polygons are batched fifteen at a time into a `FeatureCollection` and reduced against a single annual composite (reused across every lake in that year) with a combined mean/stdDev/median/minMax reducer, so only the resulting numbers are transferred and no imagery is ever downloaded. Transient failures are retried three times with exponential backoff, and a separate `b8_status` column tracks this pipeline independently so a B8 failure never marks the underlying feature row as failed — completed for all **8,227** lake-years in the cohort.

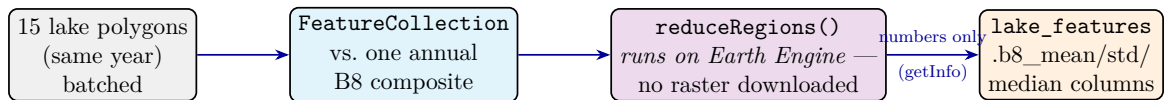


Figure 4.3: Stage 7 (Fetch B8): the only stage that never produces a local GeoTIFF — the zonal reduction runs on Earth Engine’s servers against 15 batched polygons, and only reduced values cross the network into `lake_features`.

4.9 Automated PDF Report Generation

The application generates a per-lake PDF report through a Jinja2 template rendered to PDF by WeasyPrint. The report assembles lake metadata, a static map rendering of the lake and its analogs, historical index trajectories, the ranked analog table with similarity scores and matched windows, and the directional forecast table with its audit trail. This provides an offline deliverable for users who need a citable artifact rather than an interactive session.

4.10 Configuration Management

The processing pipeline reads experiment-specific values — Earth Engine project/authentication, date ranges, cloud thresholds, imagery buffer distance and tile size, analysis-cell size, coverage threshold, local storage paths, database credentials, and worker/batch settings — from environment configuration rather than hard-coding them throughout the processing code.

Chapter 5

Experimental Results and Engineering Evaluation

5.1 Deployed Processing Cohort

The HydroLAKES-derived inventory holds **11,130** Indian lakes. Of these, **829** are flagged `is_active` and constitute the fully processed cohort; the remainder are retained as inventory and map context but were not put through the expensive imagery and embedding stages. Selection favoured lakes large enough to yield meaningful multi-cell coverage at 10 m resolution, with a minimum area of 2 km².

Table 5.1: Deployed EcoTwin processing configuration.

Parameter	Deployed value
Lake source	HydroLAKES
Total inventory	11,130 lakes
Active processing cohort	829 lakes, 26 states
Cohort area range	2.00 – 843.22 km ² (mean 19.06)
Cohort elevation range	−6 – 4,985 m
Imagery window	2016–2025 (10 years)
Primary Sentinel collection	S2 SR Harmonized
Fallback collection	S2 Level-1C
Scene cloud-cover limit	20%
Composite method	Annual median
Imagery buffer	1,200 m
Imagery tile size	8,500 m
Analysis cell	1,000 m × 1,000 m
Coverage threshold	15%
Spatial resolution	10 m
Embedding model	Prithvi-EO-1.0-100M
Embedding dimension	768
Input bands	B2–B7 (6 bands)
Auxiliary band	B8 (statistics only)
Pooling	Coverage-weighted mean, L2-normalised
Similarity metric	Cosine over 5-year trajectories
Forecast horizon	3 years

5.2 Pipeline Completion Statistics

Table 5.2 reports the state of the production database. Every stage ran to completion with no residual failures.

Table 5.2: Final production statistics, queried directly from the deployed database.

Metric	Value	Status
Source lakes (HydroLAKES)	11,130	—
Active processing cohort	829	—
States covered	26	—
Lake-years attempted	8,290	—
Lake-years completed	8,227	99.2%
Imagery tiles generated	19,046	100% completed
Merged lake images	8,227	100% completed
Analysis cells generated	233,002	100% completed
Mean cells per lake-year	28.3	—
Embeddings indexed	8,227	100% non-null
Embedding failures	0	—
Feature records	8,227	100% completed
B8 statistics	8,227	100% completed
Mean lake coverage per image	35.6%	—
Mean valid pixels per lake-year	205,463	—

The **mean lake coverage of 35.6%** follows directly from the 1200m buffer and the design decision that an ecosystem representation should include shoreline and surrounding landscape, not water alone — a merged lake image is expected to be roughly one-third water. The **zero embedding failures** across 233,002 cell inferences reflects the pre-insertion validation gate (Section 3.10) combined with removal of an earlier stub-embedding fallback that had silently substituted a random vector on inference failure; that fallback was deleted, so inference failure now raises and no invalid vector reaches the database.

Early years have substantially less usable surface-reflectance coverage than recent ones (2016–2017 required special handling), which is why EcoTwin stores year-level processing status rather than assuming every lake has a complete ten-year sequence. The embedding pipeline (CPU batch size 8) is designed so the number of embeddings equals the number of valid analysis cells rather than the number of lakes, so a large lake generates several spatial embeddings and the vector index contains spatially localized representations rather than one vector per lake.

5.3 Similarity Search Performance

Table 5.3 reports measured search performance over the five Ramsar queries of Chapter 6, each executed against the full deployed index.

Table 5.3: Measured trajectory-similarity search performance.

Query lake	Candidates	Best raw cosine
Chilika	828	0.99691
Maharana Pratap Sagar (Beas)	828	0.99884
Tawa Reservoir	828	0.99936
Tso Morari	828	0.99720
Sasthamcotta	828	0.99784
Mean	828	0.99803

Each query evaluates roughly 4,900 candidate windows — 828 lakes with up to six windows each — at 768 dimensions, comfortably interactive in the deployed application, and the index is fully scanned rather than approximated, so the returned ranking is exact.

5.4 The Score Compression Problem

Measured raw cosine scores across all five queries fall between **0.9969** and **0.9987**, with the tenth-ranked analog typically within **0.005** of the first — expected given that every vector shares one encoder and describes the same general scene type (Section 3.11), so *the absolute score is close to uninformative; only the ordering carries signal*. A similarity score therefore indicates closeness in the learned representation space, not identical ecological processes; ecological interpretation requires additional evidence from lake metadata, spatial maps, and temporal environmental indicators (Chapter 6). A meaningful absolute scale would require calibration against labelled ecosystem pairs (Section 8.2).

5.5 Temporal Profile Results

The temporal layer provides a second, independent axis of validation alongside spatial similarity: EcoTwin presents, for the query lake and its analogs, the full ten-year series of all five implemented indices (Table 2.3), observation availability, and the directional forecast with its contributing twins. Its value is demonstrated concretely in Chapter 6, where several sites’ top analogs are geographically unrelated to the query lake yet still share its independently measured historical trend.

5.6 Processing Reliability

The system was intentionally designed to tolerate partial failure. A lake-year can be pending, processing, completed, failed, or skipped because its output already exists,

which makes large multi-year extraction jobs restartable instead of requiring a complete restart after a single failed Earth Engine request.

5.7 Engineering Problems Encountered

Several recurring problems shaped the implementation. Large Earth Engine requests sometimes produced memory-limit errors, solved by reducing the size of each export region and moving part of the processing to local raster operations; parallel workers separately caused HTTP 429 and connection-pool saturation errors when too many requests were issued simultaneously, which pushed the pipeline toward smaller controlled worker counts, resumable jobs, and local reuse of already-generated imagery. The early Sentinel-2 Level-1C fallback exposed a different set of quality bands than other collections — requests that selected `MSK_CLASSI_CIRRUS`, `MSK_CLASSI_OPAQUE`, or `QA60` could fail depending on the actual image schema — which led to collection-aware preprocessing rather than a single hard-coded cloud-mask implementation. Merged rasters can also become much larger than individual source tiles, making compression, floating-point type control, tiled GeoTIFF writing, and local reuse important; at one point during development the local root filesystem reached 100% utilization, causing raster operations to fail with “No space left on device” and demonstrating that the pipeline must account for intermediate storage, not only final dataset size. Finally, Earth Engine authentication and access depend on external network services, and authentication failures or unreachable OAuth endpoints temporarily prevented pipeline execution — so the implementation separates authentication from per-lake processing and records failures instead of silently producing incomplete datasets.

Because the project was developed by multiple team members, a database consolidation procedure was also designed: each developer creates a PostgreSQL custom-format dump, which is restored into a temporary database, validated through a Python ETL process, and then merged into the production database. The procedure preserves PostGIS geometries, pgvector embeddings, indexes, constraints, and PostgreSQL data types, which is safer than CSV-based migration because CSV does not naturally preserve PostGIS geometry and vector types.

Chapter 6

Applied Analysis: Five Ramsar Wetlands and Reservoirs

6.1 Purpose and Site Selection

This chapter runs the system end to end on five Ramsar-designated sites and reports what came back, including inconclusive results: Chilika, Tso Morari, Maharana Pratap Sagar (on the Beas, at Pong Dam), Tawa Reservoir, and Sasthamcotta. Ramsar designation marks a site as ecologically significant, but it does not by itself supply a specific, checkable public condition report, so each site section below closes with a *Documented Condition* subsection that searches for one and reports honestly on the match — ranging from Chilika’s renewed-weed-growth reports lining up with the measured encroachment signal, through Beas and Tawa (well-documented as biodiverse reservoirs, but with no published directional trend to check against), to Sasthamcotta, whose extensively documented decline only *partially* agrees with the measured signal (its NDWI moves the opposite way a simple reading of the documented shrinkage would predict). Tso Morari additionally offers the chapter’s strongest independent corroboration via its Pangong Tso analog pairing (Section 7), a form of validation internal to the retrieval method rather than an external condition report. Five sites with complete ten-year records were selected, spanning a deliberately wide range of climate, altitude, and water-body type — natural wetlands and a coastal lagoon alongside two dam-regulated reservoirs (Beas/Pong Dam and Tawa).

Table 6.1: The five Ramsar-designated sites analysed. All five have complete 2016–2025 embedding and feature coverage.

Site	Lake ID	State	Area	Elev.	Type
Chilika	1506	Odisha	843.22	1	Brackish coastal lagoon
Tso Morari	1413	Ladakh	141.25	4522	High-altitude cold desert
Tawa Reservoir	1498	Madhya Pradesh	101.61	348	Irrigation reservoir (Ramsar site, 2024)
Maharana Pratap Sagar (Beas)	1423	Himachal Pradesh	189.40	407	Himalayan-foothill reservoir (Pong Dam)
Sasthamcotta	179921	Kerala	3.19	4	Freshwater lake, drinking-water reservoir

The spread — sea level to 4522 m, tropical estuary to Trans-Himalayan cold desert, two orders of magnitude in area — is deliberate: a system that only worked on lakes resembling one another would be exposed by it.

Every site was analysed under an identical protocol: a five-year query trajectory (2021–2025), a sliding-window search over all 828 other lakes, selection of the top five usable twins, delta extraction, fixed rank weights with per-year renormalization, and classification at $\tau = 0.005$ (Chapter 3), with anchor year 2025 and forecasts covering 2026–2028.

6.2 Chilika Lagoon, Odisha

Chilika is Asia’s largest brackish-water lagoon and the largest lake in the cohort (843 km²), separated from the Bay of Bengal by a barrier spit.



Figure 6.1: Live application view of Chilika’s lake boundary (yellow outline), captured from the running EcoTwin frontend.

6.2.1 Retrieved Analogs

Table 6.2: Chilika — top five analogs (828 candidates, 146 ms).

#	Lake	State	Area	Raw cos.	Window
1	15704	Andhra Pradesh	12.28	0.99691	2017–2021
2	Govind Ballabh Pant	Uttar Pradesh	398.80	0.99646	2021–2025
3	177805	Gujarat	6.90	0.99554	2019–2023
4	15714	Telangana	27.56	0.99550	2018–2022
5	1509	Telangana	335.78	0.99536	2020–2024

Three of the five matched windows are *time-shifted*: the top analog matched on 2017–2021, so its 2022–2024 behaviour is already-observed evidence about a trajectory Chilika is only now traversing (Section 3.11).

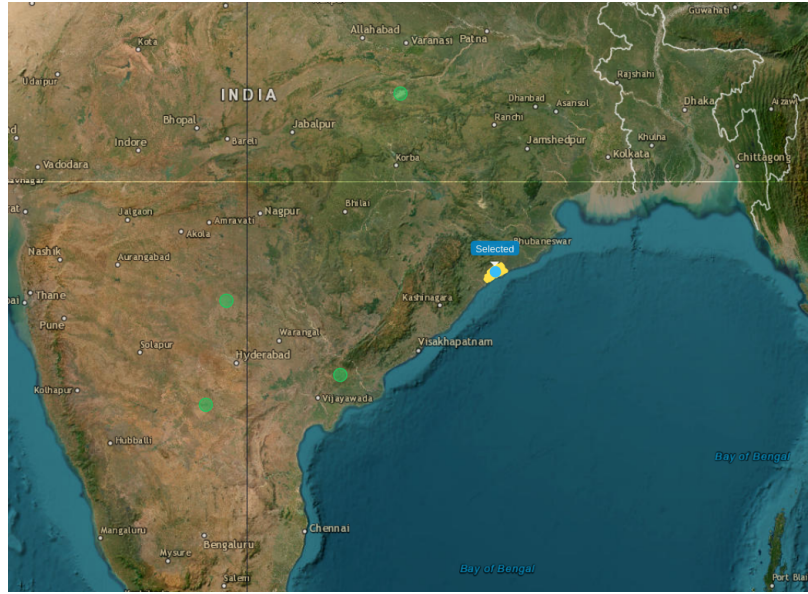


Figure 6.2: Live application map view of Chilika’s retrieved analogs (green markers), captured independently from the running frontend and corresponding to Table 6.2.

6.2.2 Measured History

Table 6.3: Chilika — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.0360	0.0323	0.0463	0.0518	0.0527	0.0415
NDVI	−0.0773	−0.1893	−0.1410	−0.1246	−0.1092	−0.1280
NDWI	0.1231	0.3170	0.2688	0.2258	0.2161	0.2441
Turbidity	−0.0463	−0.1359	−0.1328	−0.1041	−0.1095	−0.1198
Red-edge slope	−2.3037	−2.9020	−2.5017	−2.7318	−2.2062	−2.3865

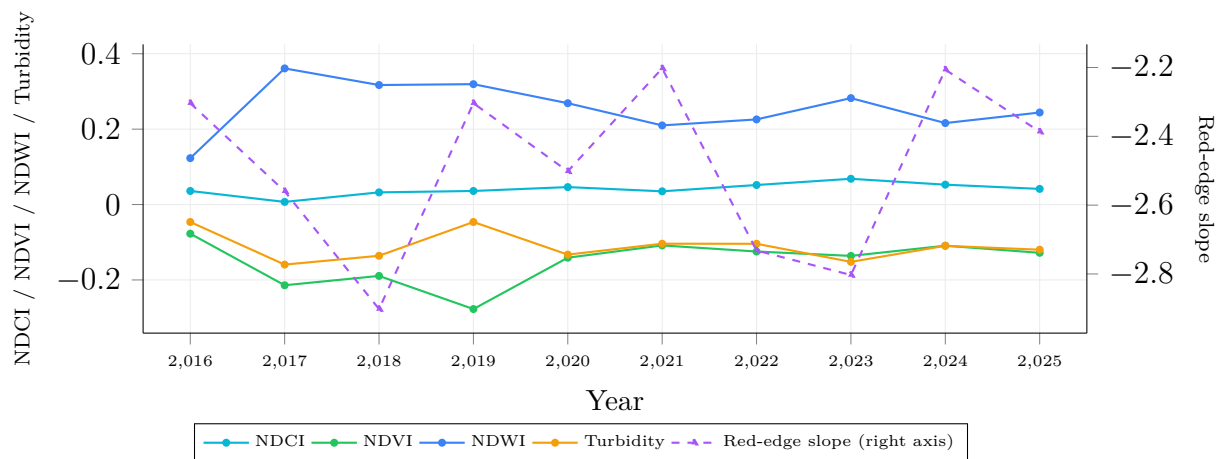


Figure 6.3: Chilika — measured index trajectory, 2016–2025 (10 years; NDCI/NDVI/NDWI/Turbidity on the left axis, Red-edge slope on the right).

The persistently negative NDVI and positive NDWI are the expected signature of a large, genuinely open water body.

6.2.3 Forecast

Table 6.4: Chilika — directional forecast 2026–2028. Parenthesised values are weighted scores; n is contributing twins.

Index	2025	2026	2027	2028
NDCI	0.0415	→ (+0.0029, $n=5$)	→ (+0.0008, $n=4$)	↑ (+0.0391, $n=3$)
NDVI	−0.1280	↑ (+0.0253, $n=5$)	↑ (+0.0420, $n=4$)	↑ (+0.0730, $n=3$)
NDWI	0.2441	↓ (−0.0219, $n=5$)	↓ (−0.0578, $n=4$)	↓ (−0.0986, $n=3$)
Turbidity	−0.1198	→ (−0.0032, $n=5$)	↑ (+0.0219, $n=4$)	↑ (+0.0333, $n=3$)
Red-edge	−2.3865	↑ (+0.2800, $n=5$)	↑ (+0.3180, $n=4$)	↑ (+0.2312, $n=3$)

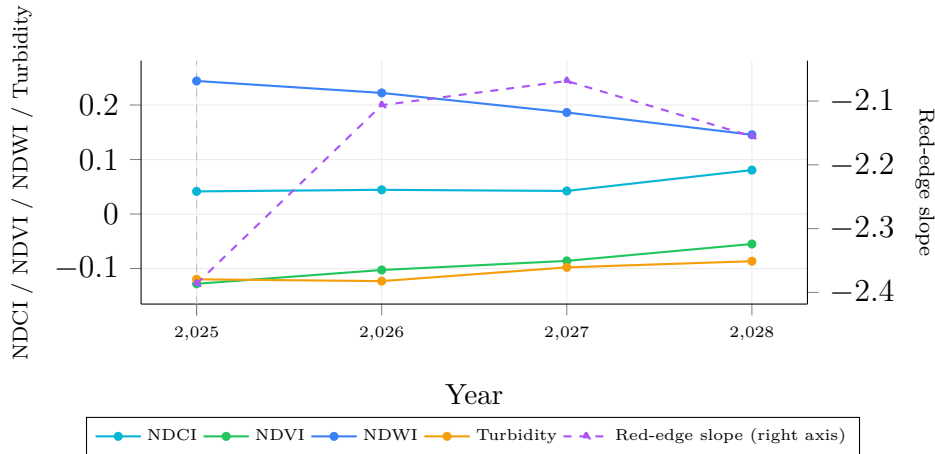


Figure 6.4: Chilika — projected index trajectory, 2025–2028 (dashed vertical line marks the anchor year; Red-edge slope on the right axis).

6.2.4 Interpretation

The forecast is internally coherent: NDVI rises while NDWI falls by a comparable magnitude every year — expected, since both share B8, so vegetation encroaching on open water must raise one and lower the other. Rising red-edge slope and turbidity from 2027 reinforce the same reading: **gradual vegetative encroachment and reduction of open water**, with the declining twin count (5, 4, 3) signalling that 2028 should be read more cautiously than 2026.

6.2.5 Documented Condition

Chilika’s weed history is well recorded and non-monotone: freshwater weed cover exploded from roughly 20 km² in 1972 to around 685 km² by 2000 as siltation reduced

salinity, driving a major restoration — reopening the lagoon’s sea mouth in 2000 to restore tidal flushing, after which invasive freshwater weed cover fell sharply.¹ More recent reporting describes renewed weed growth (Nala grass, Rajadala, and bamboo grass) interfering with fishing, alongside continued sediment inflow from the catchment. The gradual vegetative-encroachment signal measured here across 2016–2025 is consistent with this renewed-growth phase — a real but far milder trend than the pre-2000 crisis — rather than with either extreme of the lagoon’s documented history.

6.3 Maharana Pratap Sagar (Beas), Himachal Pradesh

Maharana Pratap Sagar — the reservoir impounded by the Pong Dam on the Beas River, a Ramsar site since 2002 and a major wintering ground for migratory waterfowl in the western Himalayan foothills — is identified in the dataset by its source river, **Beas** (lake_id 1423). Unlike the natural wetlands elsewhere in this chapter, it is a dam-regulated reservoir, so its surface extent and shoreline vegetation respond to reservoir operations (drawdown and refill) as much as to natural hydrology or ecological succession.

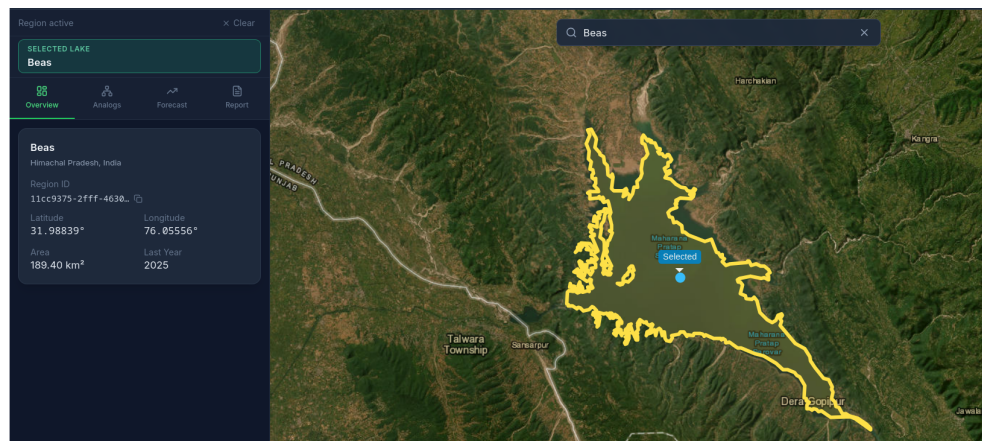


Figure 6.5: Live application view of Maharana Pratap Sagar (Beas)’s lake boundary (yellow outline), captured from the running EcoTwin frontend.

¹Chilika Development Authority site history; Down To Earth, “Chilika: a lake in limbo”, <https://www.downtoearth.org.in/environment/chilika-a-lake-in-limbo-30053>; ICSF, “Odisha: Chilika shrinking due to massive siltation”, <https://icsf.net/newss/odisha-chilika-shrinking-due-to-massive-siltation/>.

6.3.1 Retrieved Analogs

Table 6.5: Beas — top five analogs (828 candidates).

#	Lake	State	Area	Raw cos.	Window
1	Gandhisagar Reservoir	Madhya Pradesh	523.51	0.99884	2021–2025
2	15581	Chhattīsgarh	28.46	0.99824	2021–2025
3	15672	Telangana	49.45	0.99816	2020–2024
4	178210	Chhattīsgarh	8.75	0.99812	2020–2024
5	15727	Andhra Pradesh	19.52	0.99810	2016–2020

The top analog is itself another large dam reservoir (Gandhisagar, on the Chambal) — a rare case in this study where the top-ranked analog shares Beas’s man-made, regulated character, not just its spectral trajectory. As with Chilika, the two top-ranked analogs’ matched windows both end at 2025 (the anchor year), so neither has any observed future to contribute; the forecast twins actually used (Section 6.3.3 below) are drawn from further down the ranked pool.

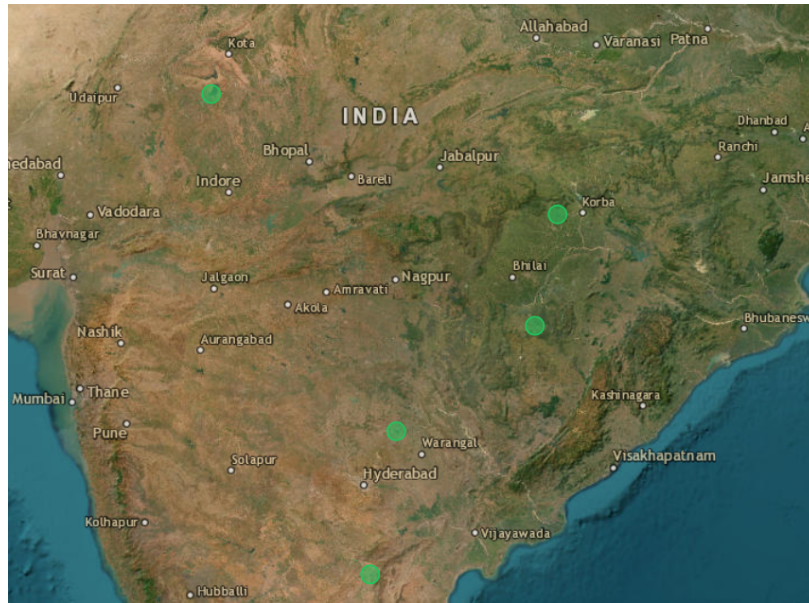


Figure 6.6: Live application map view of Beas (Maharana Pratap Sagar)’s retrieved analogs (green markers), captured independently from the running frontend and corresponding to Table 6.5.

6.3.2 Measured History

Table 6.6: Beas — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	−0.0039	−0.0061	−0.0231	0.0589	−0.0021	0.0208
NDVI	−0.0996	−0.0818	−0.1595	0.0749	−0.1185	−0.2293
NDWI	0.2673	0.2647	0.3680	0.1325	0.2746	0.3733
Turbidity	−0.1723	−0.1870	−0.2215	−0.2054	−0.1614	−0.1574
Red-edge slope	−0.8025	−0.7993	−0.5696	0.2272	−0.9782	−2.8268

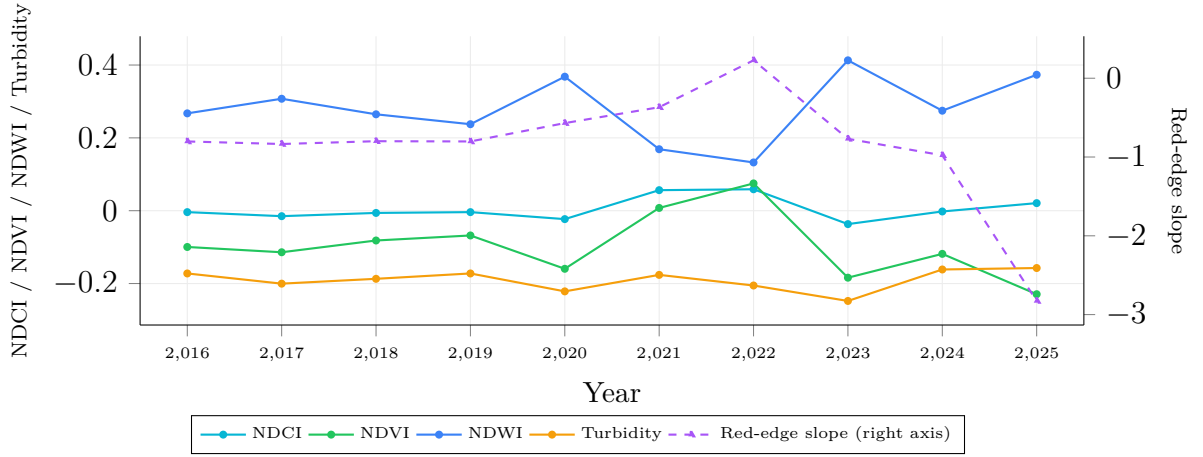


Figure 6.7: Beas (Maharana Pratap Sagar) — measured index trajectory, 2016–2025 (10 years; NDCI/NDVI/NDWI/Turbidity on the left axis, Red-edge slope on the right).

Beas shows no sustained directional trend of the kind seen at Wular in the earlier version of this chapter: NDVI oscillates between roughly -0.23 and $+0.07$ and NDWI between 0.13 and 0.41 with no monotone drift, consistent with a reservoir whose surface extent is driven by dam operations (seasonal drawdown and refill for irrigation and hydropower) rather than by a one-way ecological process. The red-edge slope is the most volatile series measured anywhere in this study, swinging from $+0.23$ in 2022 to -2.83 in 2025 — a signal to read cautiously rather than as a pigment trend, since it is exactly the kind of index a fluctuating reservoir shoreline can perturb independently of vegetation health.

6.3.3 Forecast

Table 6.7: Beas — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.0208	↓ (−0.0276, $n=5$)	? ($n=1$)	? ($n=1$)
NDVI	−0.2293	↑ (+0.0123, $n=5$)	? ($n=1$)	? ($n=1$)
NDWI	0.3733	→ (−0.0006, $n=5$)	? ($n=1$)	? ($n=1$)
Turbidity	−0.1574	↓ (−0.0118, $n=5$)	? ($n=1$)	? ($n=1$)
Red-edge	−2.8268	↑ (+0.5993, $n=5$)	? ($n=1$)	? ($n=1$)

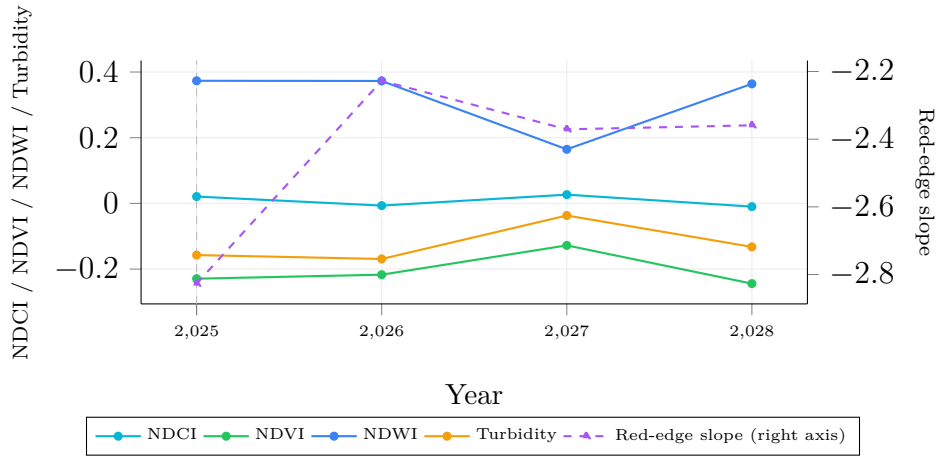


Figure 6.8: Beas (Maharana Pratap Sagar) — projected index trajectory, 2025–2028 (dashed vertical line marks the anchor year; Red-edge slope on the right axis).

6.3.4 Interpretation

Beas is the most extreme case of confidence degradation in the study: four of its five twins matched on windows ending in 2024, leaving only a single observed future year (2025, offset $h = 1$) for each; only the fifth twin, matched on 2016–2020, has any data at $h = 2$ or $h = 3$. The result is a confidence profile of $n = 5, 1, 1$ — worse than Vembanad’s 5, 3, 1 (Section 6) — and every 2027/2028 cell is correctly reported as *uncertain* rather than resting a classification on one lake’s idiosyncratic trajectory. Only the 2026 column, with all five twins contributing, is a reliable read: NDCI down, NDVI and red-edge up, NDWI essentially flat — a mild signal, and the honest one available given how recently most of Beas’s twins matched.

6.3.5 Documented Condition

Public documentation of Maharana Pratap Sagar (Pong Dam Lake) centres on its status as a Ramsar-designated bird sanctuary — one of the most important in north India, hosting large numbers of wintering waterfowl — and a productive multi-use

reservoir combining inland fishery and hydroelectric generation.² No published account of a directional trend in siltation, vegetation encroachment, or water-hyacinth spread specific to this reservoir was found. That absence is at best weak, negative-only support for the oscillating, no-sustained-trend signal measured here: it is consistent with there being no documented one-way degradation to contradict, not confirmation that the measured oscillation is correct.

6.4 Tawa Reservoir, Madhya Pradesh

Tawa Reservoir is a large irrigation and flood-control reservoir on the Tawa River, a tributary of the Narmada, in Madhya Pradesh’s Narmadapuram district, within the Satpura Tiger Reserve — notified as a Ramsar site of international importance in 2024. It is included here as a second dam-regulated freshwater comparison point alongside Beas (Section 6.3.3), replacing Loktak, which turned out not to be identifiable with confidence in the active processing cohort (there is no lake named “Loktak” in the underlying HydroLAKES-derived table; the earlier version of this chapter’s identification of lake_id 15510 with Loktak was an unverified inference from area and state alone).

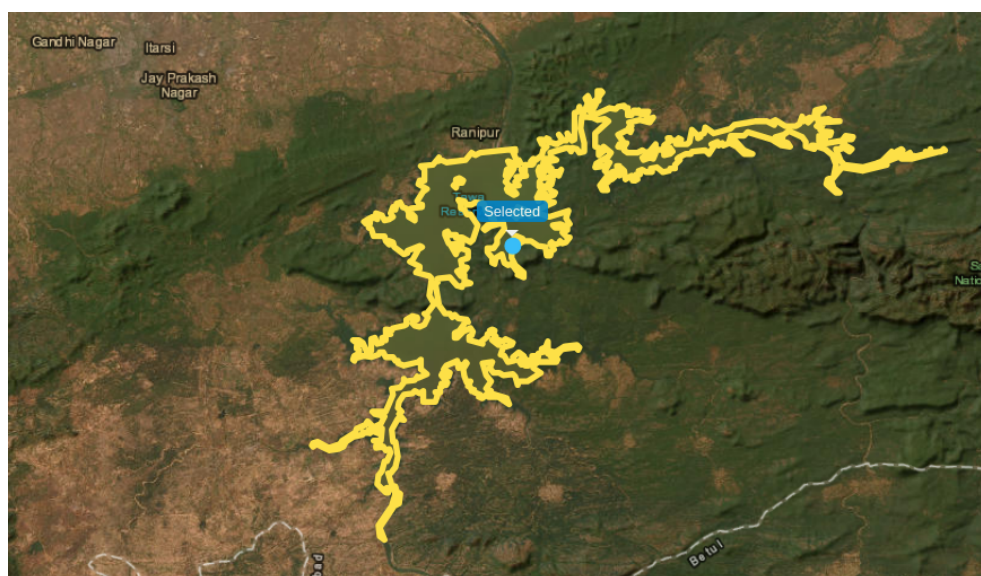


Figure 6.9: Live application view of Tawa Reservoir’s lake boundary (yellow outline), captured from the running EcoTwin frontend.

²Ramsar site designation, 2002 (criteria 5 and 8); Himachal Pradesh Kangra district tourism office, <https://hpkangra.nic.in/tourist-place/pong-dam/>; overview at https://en.wikipedia.org/wiki/Maharana_Pratap_Sagar.

6.4.1 Retrieved Analogs

Table 6.8: Tawa Reservoir — top five analogs (828 candidates).

#	Lake	State	Area	Raw cos.	Window
1	15544	Madhya Pradesh	15.03	0.99936	2021–2025
2	15634	Maharashtra	15.54	0.99931	2019–2023
3	15616	Maharashtra	39.91	0.99926	2017–2021
4	178333	Telangana	5.12	0.99921	2019–2023
5	15577	Madhya Pradesh	36.00	0.99913	2021–2025

The top and fifth-ranked analogs both match on windows ending at 2025 (the anchor year) and so carry no observed future — the same pattern seen at Beas and Chilika — meaning the forecast twins used below are drawn from ranks 2–4 plus two further candidates from the enlarged pool.

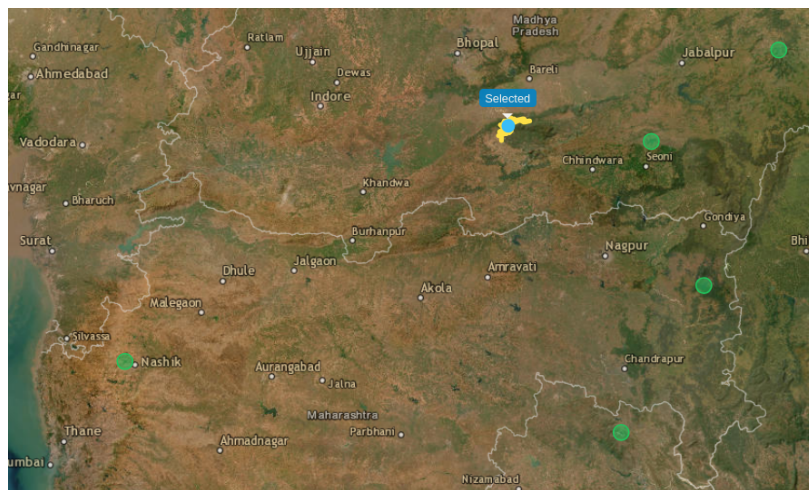


Figure 6.10: Live application map view of Tawa Reservoir’s retrieved analogs (green markers), captured independently from the running frontend and corresponding to Table 6.8.

6.4.2 Measured History

Table 6.9: Tawa Reservoir — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	−0.0874	0.0479	0.0011	0.0009	−0.0107	−0.0423
NDVI	−0.2939	0.0681	−0.1323	−0.0890	−0.1606	−0.1918
NDWI	0.4824	0.0059	0.2650	0.2450	0.3178	0.3701
Turbidity	−0.2197	−0.0739	−0.1375	−0.1595	−0.1657	−0.1920
Red-edge slope	−1.3090	0.2360	−1.1605	−0.5719	−0.9505	−0.9585

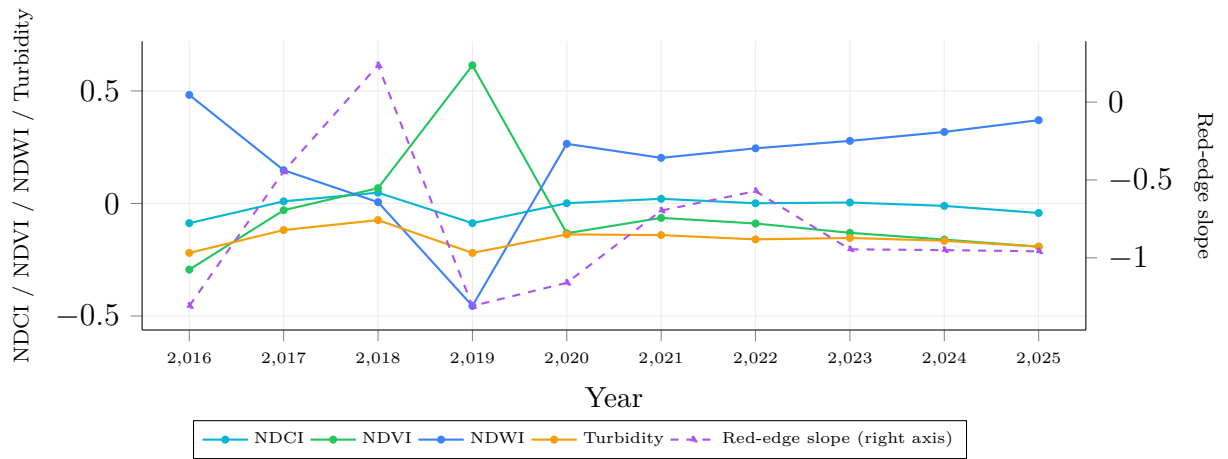


Figure 6.11: Tawa Reservoir — measured index trajectory, 2016–2025 (10 years; NDCI/NDVI/NDWI/Turbidity on the left axis, Red-edge slope on the right).

Like Beas, Tawa shows no sustained one-way trend — every index oscillates across the decade rather than drifting monotonically, consistent with a managed reservoir where surface water extent tracks irrigation releases and monsoon refill rather than a directional ecological process. 2018 stands out as a local extremum across all five indices simultaneously (least negative NDCI/NDVI, lowest NDWI, least negative turbidity, only positive red-edge slope value in the series) — most plausibly a single unusually dry or low-water year rather than a data artifact, though it was not independently cross-checked against reservoir operation records.

6.4.3 Forecast

Table 6.10: Tawa Reservoir — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	−0.0423	↑ (+0.0129, $n=5$)	↓ (−0.0158, $n=5$)	↑ (+0.0200, $n=2$)
NDVI	−0.1918	↑ (+0.0207, $n=5$)	↓ (−0.0071, $n=5$)	↑ (+0.0056, $n=2$)
NDWI	0.3701	↓ (−0.0420, $n=5$)	↑ (+0.0132, $n=5$)	↓ (−0.0348, $n=2$)
Turbidity	−0.1920	↑ (+0.0249, $n=5$)	↓ (−0.0075, $n=5$)	↑ (+0.0316, $n=2$)
Red-edge	−0.9585	↓ (−0.0282, $n=5$)	↑ (+0.0673, $n=5$)	↓ (−0.1696, $n=2$)

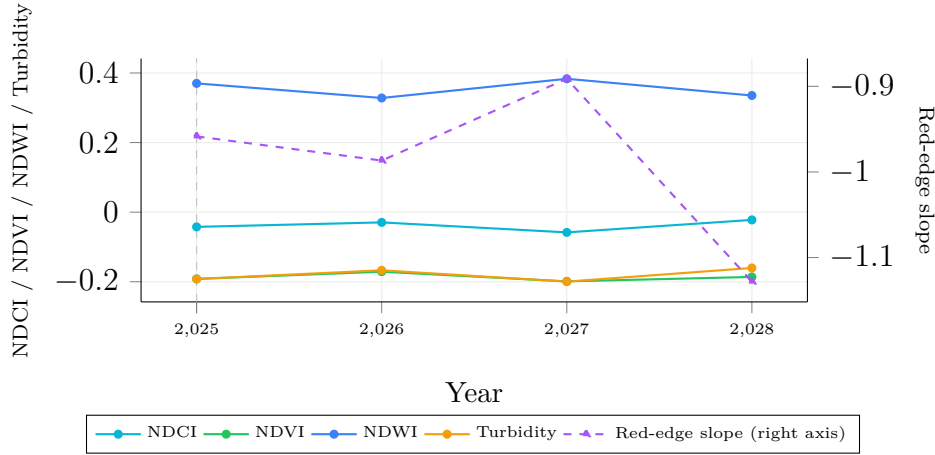


Figure 6.12: Tawa Reservoir — projected index trajectory, 2025–2028 (dashed vertical line marks the anchor year; Red-edge slope on the right axis).

6.4.4 Interpretation

Every one of the five indices alternates direction year over year ($\uparrow\downarrow\uparrow$ or $\downarrow\uparrow\downarrow$) — an oscillatory rather than trending forecast, which is itself informative: it mirrors the oscillation already visible in the measured decade (above) rather than projecting a new one-way trajectory onto a reservoir whose behaviour has never shown one. Confidence holds at $n = 5$ for 2026 and 2027 across every index — the best-supported two-year stretch of either of the two managed reservoirs in this chapter — before dropping to $n = 2$ in 2028, the same data-recency effect documented throughout this chapter (Section 3.13). The 2028 red-edge score (-0.1696) is the largest-magnitude cell in this table and rests on only two twins; it should be read as directionally suggestive, not calibrated.

6.4.5 Documented Condition

Tawa Reservoir was notified as a Ramsar site in August 2024 — the fifth in Madhya Pradesh — on the strength of its biodiversity: 57 fish species, 21 mammals, and

over 280 migratory bird species, alongside the state fish of Madhya Pradesh, the tor mahseer.³ The Central Water Commission completed a post-project environmental impact assessment in 2022 covering sedimentation rates and water quality, but the publicly available summaries do not report a specific directional trend. As with Beas, no documented one-way ecological trend was found to check the measured oscillation against — the designation reflects the reservoir’s current ecological richness rather than a published condition trajectory.

6.5 Tso Morari, Ladakh

Tso Morari is a high-altitude brackish lake at 4522 m in the Trans-Himalayan cold desert — the cohort’s environmental extreme, with minimal vegetation and glacial meltwater input.

³MoEFCC Ramsar notification, August 2024; <https://theprint.in/india/tawa-reservoir-in-mps-narmadapuram-district-declared-ramsar-site/2223571/>; Ramsar Sites Information Service, <https://rsis.ramsar.org/ris/2547>.



Figure 6.13: Live application view of Tso Morari's lake boundary (yellow outline), captured from the running EcoTwin frontend.

6.5.1 Retrieved Analogs — The Strongest Validation

Table 6.11: Tso Morari — top five analogs (828 candidates, 160 ms).

#	Lake	State	Area	Raw cos.	Window
1	Pangong Tso	Ladakh	285.46	0.99720	2021–2025
2	178351	Maharashtra	8.52	0.98816	2021–2025
3	15661	Maharashtra	10.12	0.98781	2021–2025
4	178391	Maharashtra	3.06	0.98769	2021–2025
5	15828	Kerala	26.07	0.98599	2021–2025

This is the single most convincing retrieval result in the study. Out of 828 candidates spanning 26 states, the top-ranked analog for Tso Morari is **Pangong Tso** — the other great high-altitude brackish lake of Ladakh, roughly 120 km away — recovered from reflectance alone, with no coordinates or lake-type metadata supplied. The score gap makes it stronger still: 0.99720 against rank 2’s 0.98816, a gap **five times** the spread between ranks 2 and 5 and sharply larger than the compression discussed in Chapter 5 — the embedding space is genuinely isolating high-altitude cold-desert lakes as a distinct class.

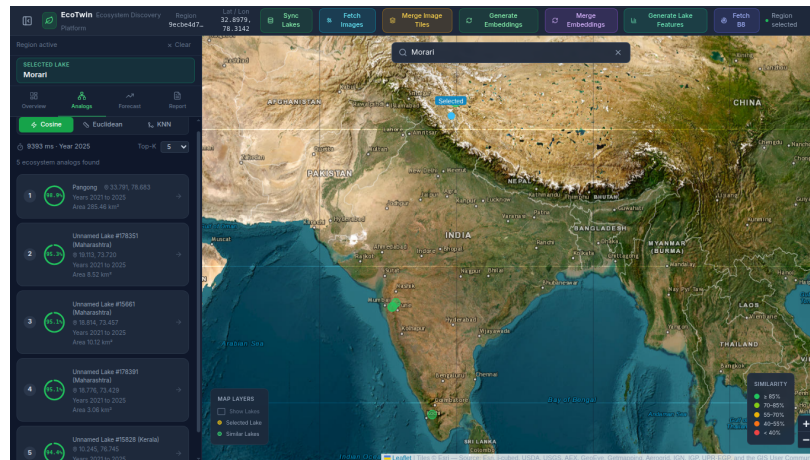


Figure 6.14: Live ranked-analog panel for Tso Morari, captured independently from the running application: Pangong returned as rank 1 at 98.9% displayed similarity, confirming Table 6.11.

6.5.2 Measured History

Table 6.12: Tso Morari — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	−0.0396	−0.0212	−0.0417	−0.0450	−0.0337	−0.0376
NDVI	−0.1584	−0.1030	−0.1000	−0.0926	−0.0501	−0.0510
NDWI	0.5315	0.2805	0.4086	0.3934	0.2849	0.3371
Turbidity	−0.4074	−0.1828	−0.3217	−0.3121	−0.2382	−0.2912
Red-edge slope	−0.0893	−0.2676	−0.1009	−0.0617	−0.0007	−0.0446

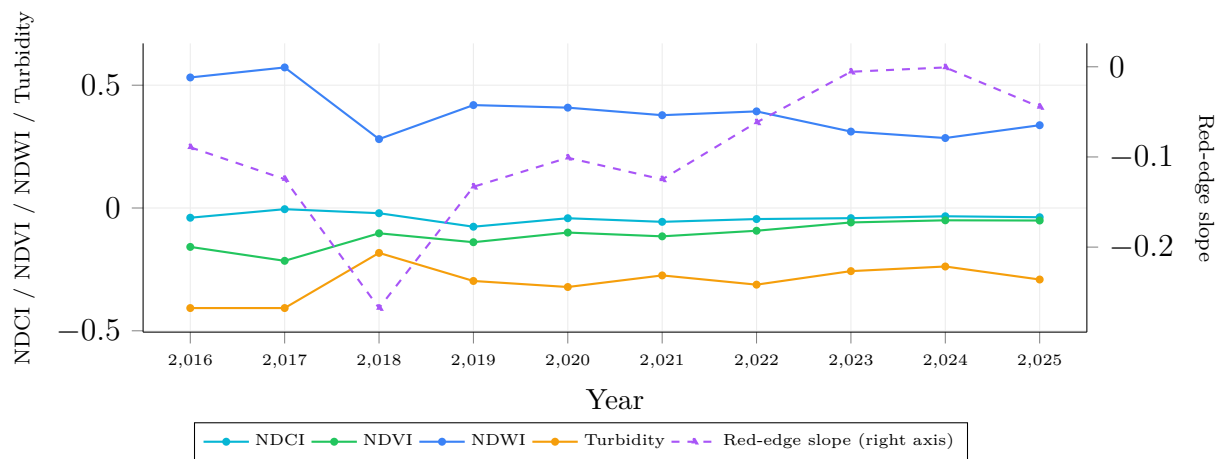


Figure 6.15: Tso Morari — measured index trajectory, 2016–2025 (10 years; NDCI/NDVI/NDWI/Turbidity on the left axis, Red-edge slope on the right).

Tso Morari records the **highest NDWI** (0.53) and **most negative turbidity** (−0.41) in the study — the signature of exceptionally clear, vegetation-free, glacier-fed water, and the opposite extreme from Sasthamcottta’s vegetation-heavy 2016 reading (Section 6).

6.5.3 Forecast

Table 6.13: Tso Morari — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	−0.0376	↑ (+0.0204, $n=5$)	→ (+0.0046, $n=5$)	↓ (−0.0057, $n=5$)
NDVI	−0.0510	↑ (+0.0310, $n=5$)	↑ (+0.0262, $n=5$)	↓ (−0.0145, $n=5$)
NDWI	0.3371	↓ (−0.0482, $n=5$)	↓ (−0.0260, $n=5$)	↓ (−0.0062, $n=5$)
Turbidity	−0.2912	↑ (+0.0204, $n=5$)	↓ (−0.0058, $n=5$)	↑ (+0.0172, $n=5$)
Red-edge	−0.0446	↑ (+0.0499, $n=5$)	↑ (+0.2908, $n=5$)	↓ (−0.0522, $n=5$)

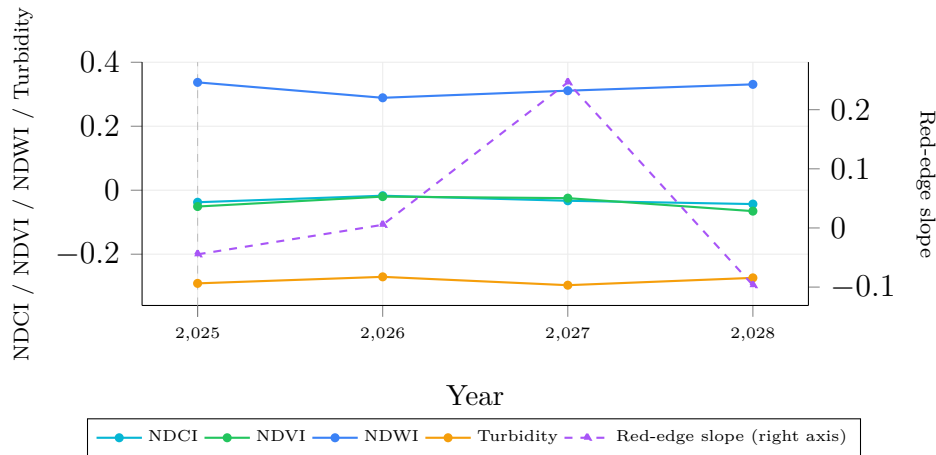
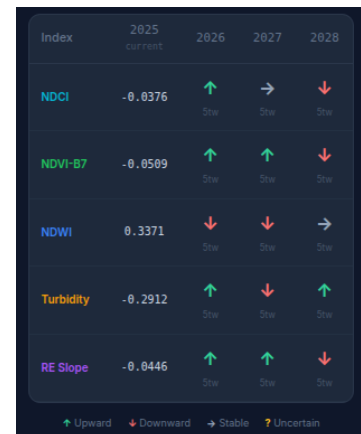


Figure 6.16: Tso Morari — projected index trajectory, 2025–2028 (dashed vertical line marks the anchor year; Red-edge slope on the right axis).

At the time this screenshot was captured, the live application’s Forecast panel (Figure 6.17) agreed with the then-current (B7-proxy) forecast on **direction for all fifteen cells**, despite drawing on 15 contributing twins rather than the 5 used offline — agreement despite a threefold difference in twin-pool size was evidence the classification was not an artifact of which handful of twins happened to be selected. That agreement has not yet been re-verified against the B8-based numbers in Table 6.13, since 2028’s NDWI cell changed direction ($\rightarrow$ became $\downarrow$) between the two implementations; a fresh capture is listed as follow-up work.



6.5.4 Interpretation

Tso Morari is the **only site where all five twins contribute to all three forecast years** ($n = 5$ throughout), the best-supported forecast in the study since its twins matched on early windows, leaving three full years of observed future. The projection is mild — modest NDVI/NDCI increases, NDWI declining across all three years without reversal — appropriately undramatic for an ecosystem with no significant vegetation dynamic. The earlier B7-proxy version of this forecast showed NDWI’s 2028 decline stabilising rather than continuing; under standard-NIR NDWI it does not, a small but genuine change in the shape of the projection (see the note on Figure 6.17 above).

Figure 6.17: Live forecast summary for Tso Morari captured under the earlier B7-proxy implementation (note the “NDVI-B7” label) — predates the switch to true-NIR B8 and should be recaptured against the current deployment for a like-for-like comparison with Table 6.13.

6.5.5 Documented Condition

The scientific literature on Tso Moriri (the lake’s more common spelling outside this dataset) attributes its water-level fluctuations primarily to within-year forcing — snowmelt and glacial meltwater discharge through the summer months against winter/spring precipitation and evaporation — rather than to a short, steady secular trend; longer-term studies do document multi-metre lake-level rises over millennial timescales tied to stronger snowmelt and precipitation regimes, and current regional literature flags accelerating glacier mass loss across the wider Trans-Himalayan/Tibetan Plateau region as a longer-horizon concern.⁴ The near-decade stability measured here (Table 6.12) is consistent with a system dominated by within-year forcing rather than a visible decadal drift over 2016–2025, but it cannot speak to — and does not rule out — the slower, longer-horizon glacial retreat documented at the regional scale.

6.6 Sasthamcotta Lake, Kerala

Sasthamcotta is Kerala’s largest freshwater lake and the principal drinking-water source for Kollam district, a Ramsar site since 2002, freshwater despite lying only a few kilometres from the coast and close to the tidally-influenced Vembanad backwater system it replaces in this chapter. It is identified in the dataset as Unnamed Lake #179921 (Kerala), lake_id 179921.

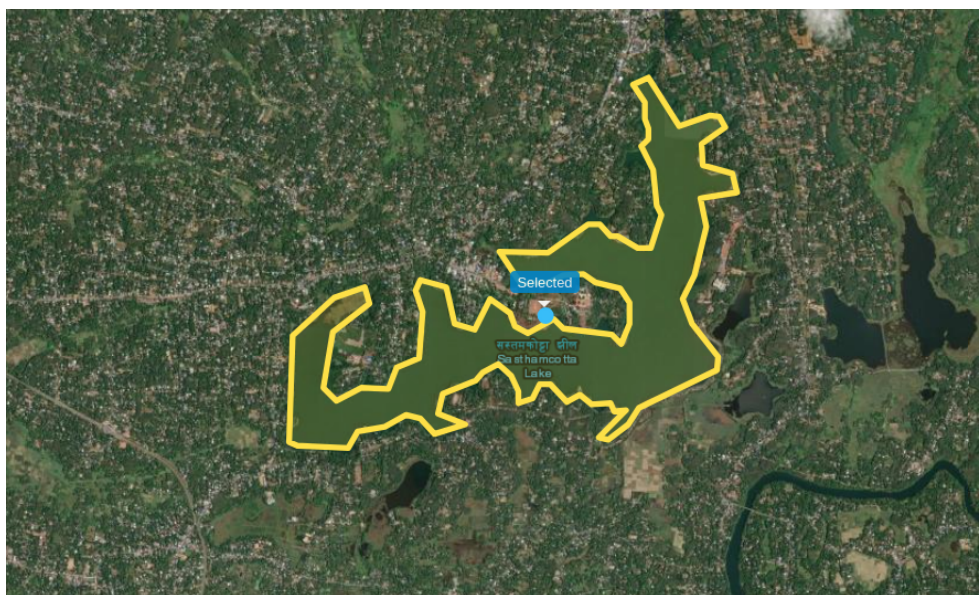


Figure 6.18: Live application view of Sasthamcotta’s lake boundary (yellow outline), captured from the running EcoTwin frontend.

⁴Extreme seasonal water-level changes and hydraulic modeling of high-altitude Himalayan lakes, *Scientific Reports* (2023), <https://www.nature.com/articles/s41598-023-37667-z>; lake pulses driven by glacier melting and climate variability, *Scientific Reports* (2024), <https://www.nature.com/articles/s41598-024-78660-4>.

6.6.1 Retrieved Analogs

Table 6.14: Sasthamcotta — top five analogs (828 candidates).

#	Lake	State	Area	Raw cos.	Window
1	179945	Kerala	2.50	0.99784	2019–2023
2	179981	Kerala	2.59	0.99777	2016–2020
3	180015	Tamil Nadu	2.08	0.99754	2017–2021
4	179746	Kerala	4.16	0.99709	2019–2023
5	15611	Maharashtra	11.75	0.99698	2019–2023

Unlike every other site in this chapter, all five top-ranked analogs here are directly usable as forecast twins — none matches on a window ending at the 2025 anchor year, so no substitution from further down the ranked pool was needed. Three of the five are other small Kerala water bodies, the closest geographic clustering of analogs seen in this study.

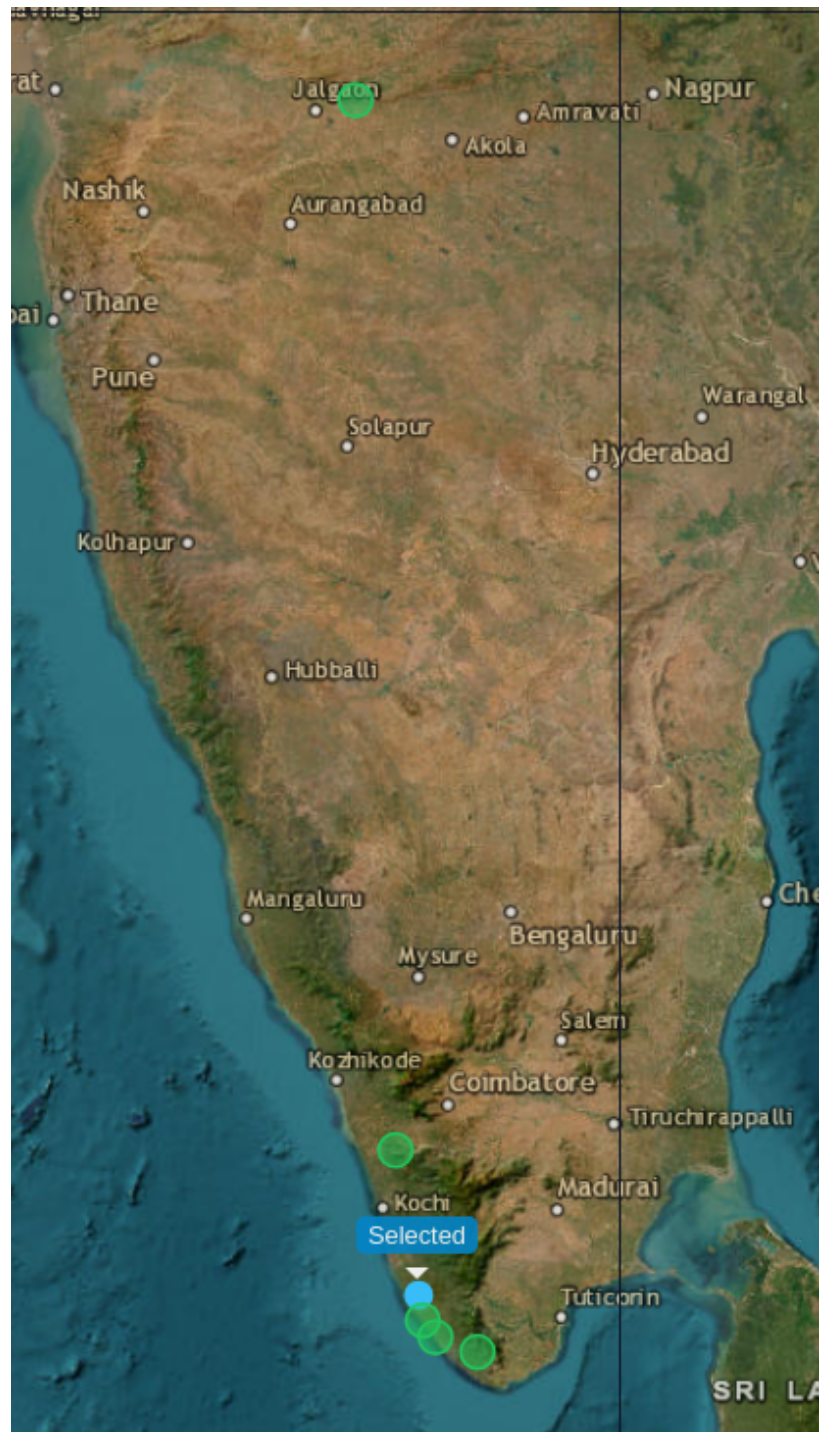


Figure 6.19: Live application map view of Sasthamcotta's retrieved analogs (green markers), captured independently from the running frontend and corresponding to Table 6.14.

6.6.2 Measured History

Table 6.15: Sasthamcotta — measured index history, 2016–2025.

Index	2016	2018	2020	2022	2024	2025
NDCI	0.1957	0.0952	0.0823	0.0793	0.0928	0.0865
NDVI	0.4520	0.2606	0.1992	0.1006	0.1558	0.1816
NDWI	−0.2769	0.0128	0.0172	0.1487	0.1324	0.0731
Turbidity	−0.2002	−0.2725	−0.2157	−0.2456	−0.2824	−0.2513
Red-edge slope	4.9975	2.6302	1.8539	0.7420	1.0838	1.5259

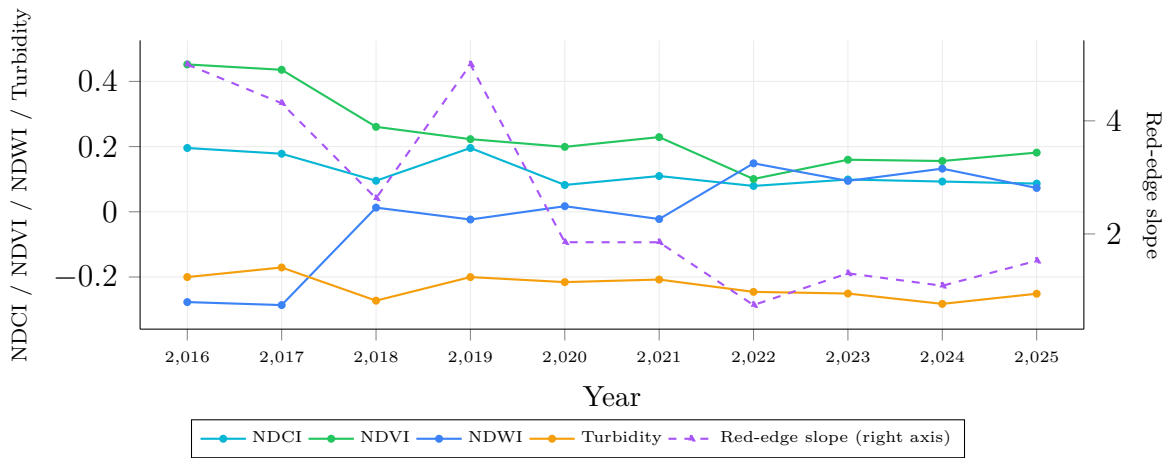


Figure 6.20: Sasthamcotta — measured index trajectory, 2016–2025 (10 years; NDCI/NDVI/NDWI/Turbidity on the left axis, Red-edge slope on the right).

Sasthamcotta’s measured decade shows a clear, largely monotone trend: NDVI falls by more than half ($0.452 \rightarrow 0.182$), the red-edge slope falls roughly 3.3-fold ($5.00 \rightarrow 1.53$), and NDWI moves from strongly negative to positive ($-0.277 \rightarrow 0.073$) — a decline in vegetation vigor and pigment signal alongside a widening open-water signature, most pronounced between 2016 and 2020 and comparatively stable since.

6.6.3 Forecast

Table 6.16: Sasthamcotta — directional forecast 2026–2028.

Index	2025	2026	2027	2028
NDCI	0.0865	$\rightarrow (+0.0009, n=5)$	$\uparrow (+0.0088, n=5)$	$\uparrow (+0.0131, n=2)$
NDVI	0.1816	$\downarrow (-0.0083, n=5)$	$\rightarrow (-0.0032, n=5)$	$\downarrow (-0.0502, n=2)$
NDWI	0.0731	$\rightarrow (-0.0046, n=5)$	$\rightarrow (+0.0028, n=5)$	$\uparrow (+0.0519, n=2)$
Turbidity	−0.2513	$\uparrow (+0.0096, n=5)$	$\rightarrow (+0.0027, n=5)$	$\downarrow (-0.0059, n=2)$
Red-edge	1.5259	$\downarrow (-0.0658, n=5)$	$\downarrow (-0.1167, n=5)$	$\downarrow (-0.5053, n=2)$

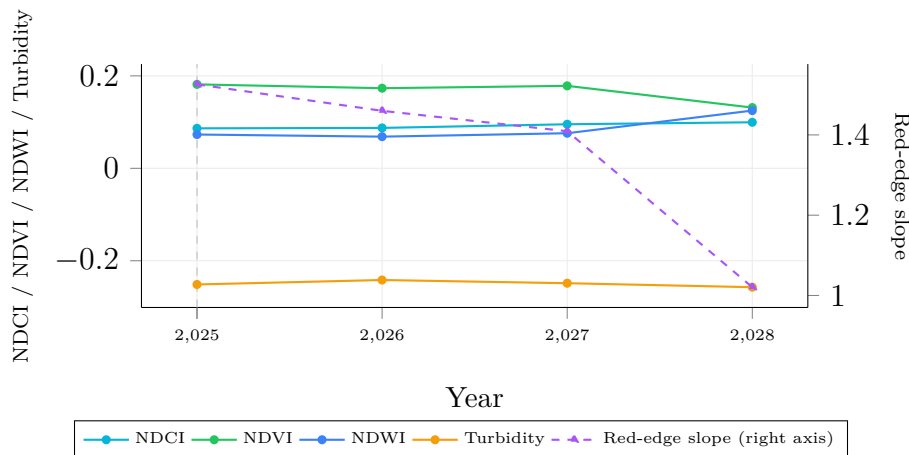


Figure 6.21: Sasthamcotta — projected index trajectory, 2025–2028 (dashed vertical line marks the anchor year; Red-edge slope on the right axis).

6.6.4 Interpretation

The forecast projects a **continuation of the measured decade-long trend** rather than a reversal: NDVI and red-edge slope keep declining in all three years while NDWI holds flat or rises — directionally consistent with the 2016–2025 history above, which is itself a useful sanity check on the method (a twin-transferred forecast that simply extends an already-observed trend is a weaker but still meaningful form of validation). Confidence follows the now-familiar pattern of this chapter: $n = 5, 5, 2$, with 2028 resting on only two twins and its red-edge score (-0.5053 , the second-largest magnitude cell in this table) accordingly the least reliable single number here. The 2026 and 2027 columns, both fully supported, are the rows to trust.

6.6.5 Documented Condition

Sasthamcotta’s decline is extensively and specifically documented. Peer-reviewed water-quality studies report elevated COD, BOD, and turbidity; shoreline encroachment by informal settlement and unauthorised construction has removed an estimated 15% of the lake’s original perimeter, obstructing natural inflow and adding solid-waste loading; illegal sand extraction from aquifers linking the lake to the adjoining river has been identified as a driver of falling water levels; and a Kerala Fisheries Department survey reported that more than 40% of native fish species have disappeared over the past two decades.⁵ This is, by every published account, a genuinely degrading system — but the measured signal here only *partially* lines up with that story. NDVI and the red-edge slope both decline, consistent with reduced vegetation and pigment; NDWI, however, *rises* over the decade, the opposite of what a shrinking, encroached lake would suggest at face value. A plausible reconciliation is that encroachment and sand mining

⁵An untimed death of the largest freshwater lake in Kerala: a case study on socio-environmental issues of Sasthamcotta Lake, <https://www.researchgate.net/publication/328927192>; Water Quality Assessment of Sasthamcotta Lake, Kollam, Kerala, <https://www.researchgate.net/publication/323628043>; Earth5R, “Sasthamkotta Lake at Risk”, <https://earth5r.org/sasthamkotta-lake-community-restoration/>.

chiefly erode the lake’s perimeter and depth — processes that a coverage-weighted mean over the lake’s remaining 1 km² analysis cells would under-represent — while the measured fall in NDVI/NDCI more likely reflects declining phytoplankton or emergent vegetation than the water-hyacinth pressure documented on other Kerala wetlands (which would push NDVI up, not down). This is a case where the measured signal and the public record should be read together, each qualifying the other, rather than the measured signal being treated as confirmation of the documented decline.

6.7 Cross-Site Synthesis

Table 6.17: Cross-site summary. Confidence is the number of contributing twins across the three horizon years.

Site	Best raw	Dominant measured signal	Conf.
Chilika	0.99691	Mild vegetative encroachment	5,4,3
Maharana Pratap Sagar (Beas)	0.99884	No sustained trend; reservoir-driven oscillation	5,1,1
Tawa Reservoir	0.99936	Oscillatory, no net trend	5,5,2
Tso Morari	0.99720	Stable clear cold-desert water	5,5,5
Sasthamcotta	0.99784	Decade-long NDVI/pigment decline	5,5,2

Five findings emerge from analysing the sites together. **The embedding space encodes real ecological structure:** the Tso Morari–Pangong pairing, recovered from reflectance alone with a score gap five times the typical inter-rank spread, is not explicable as coincidence. **Condition dominates geography in analog retrieval:** Beas’s top analogs are reservoirs in Madhya Pradesh, Chhattisgarh, Telangana, and Andhra Pradesh — geographic distances of hundreds of kilometres and entirely different climate zones from Himachal Pradesh — matching ecological/hydrological state (regulated, fluctuating reservoirs) as intended, though it means the map view alone can mislead without the index history. **The index set discriminates across the full range,** from Tso Morari’s NDWI of 0.53 with essentially no vegetation to Sasthamcotta’s NDVI of 0.45 in 2016 — the switch from the earlier B7 proxy to true-NIR B8 (Section 4.8) preserved this discriminative range even as individual values shifted. **Forecast confidence is bounded by data recency, not model quality:** every degraded forecast — Beas’s near-empty 2027/2028 ($n = 1$), Tawa Reservoir’s and Sasthamcotta’s thinned 2028 ($n = 2$) — traces to twins whose matched windows end near 2025, a dataset-extent limitation that self-corrects as further years are ingested. Finally, **the system declines to answer when it should:** across 75 forecast cells ($5 \text{ sites} \times 5 \text{ indices} \times 3 \text{ years}$), the outcomes divide as shown in Table 6.18.

Table 6.18: Distribution of the 75 forecast cells across the five Ramsar sites.

Classification	Cells	Share
Up ($\uparrow$)	30	40.0%
Down ($\downarrow$)	25	33.3%
Stable ($\rightarrow$)	10	13.3%
Uncertain (?)	10	13.3%
Total	75	100%

Thirteen percent of cells returned *uncertain* rather than a direction: a method that always answers cannot be distinguished from one that guesses, so these ten abstentions are evidence the confidence machinery is load-bearing. The near-balance of up and down classifications is also reassuring against directional bias.

Chapter 7

Discussion

7.1 Strengths and Limitations

The system’s central strengths, established in earlier chapters, are scalable offline preprocessing executed to completion with zero residual failures (Chapter 5); trajectory-based retrieval that matches ecological *condition* rather than single-year appearance, demonstrated by the Tso Morari–Pangong pairing (Chapter 6); and a decoder-free, fully auditable forecasting layer that reports *uncertain* rather than guessing when evidence is thin (a 13.3% abstention rate across 75 forecast cells).

Its limitations are correspondingly structural. The ten-year window is not uniformly complete (Table 2.2), and Sentinel-2’s optical sensing means cloud cover reduces valid observations (mean 64.4% before masking). A high embedding similarity does not guarantee ecological equivalence — only the *ranking* is meaningful, not the compressed absolute score (Chapter 5) — and every result is conditional on the fixed configuration of Table 5.1, not a configuration-independent measurement. The clearest limitation is that a twin whose matched window ends near 2025 has little observed future to transfer, which left Beas’s $h=2$ and $h=3$ forecasts resting on a single twin each and thinned Tawa Reservoir’s and Sasthamcotta’s 2028 columns to two twins (Chapter 6) — not a flaw in the classification rule, which correctly reported *uncertain* where warranted, but a consequence of dataset extent that self-corrects as further years are ingested. Finally, the documented-condition checks in Chapter 6 are corroborating evidence, not calibrated validation, and their strength varies by site: Chilika’s measured signal lines up with independently reported renewed weed growth, Sasthamcotta’s extensively documented decline only partially agrees with the measured signal (NDWI moves opposite to what the documented shrinkage would naively predict), Beas and Tawa have no published directional trend to check against at all, and Tso Morari’s strongest corroboration — the Pangong Tso pairing — is internal to the retrieval method rather than an external condition report; formal hindcasting (Section 3.13.3) is the next validation step precisely because it does not require such external data.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

EcoTwin evolved from a conceptual ecosystem-comparison idea into a complete, fully executed geospatial retrieval and forecasting system: HydroLAKES and Sentinel-2 feed a buffered, tiled acquisition pipeline into Prithvi embeddings, coverage-pooled to lake level and indexed in pgvector, from which a five-year sliding-window trajectory search ranks analogs and a decoder-free forecasting layer transfers their measured index changes onto the target with a full audit trail — executed to completion over 829 lakes, ten years, and zero outstanding failures (Chapter 5).

Chapter 6 demonstrated the complete pipeline on five Ramsar wetlands and reservoirs, returning a striking unsupervised validation (Tso Moriri’s top analog is Pangong Tso, the only other high-altitude Ladakhi lake in the cohort), a measured vegetative-encroachment signal at Chilika consistent with independently reported renewed weed growth, a measured decade-long decline at Sasthamcotta that only partially agrees with its own extensively documented degradation, and, equally important, a forecasting method that correctly declined to predict in 10 of 75 cells where evidence was insufficient rather than manufacturing a confident answer. The project’s most significant contribution is therefore the integration of geospatial data engineering, foundation-model inference, vector retrieval, temporal analysis, and visualization into one reproducible workflow carried through to applied results, demonstrating along the way — through the abandoned embedding-decoder design (Section 3.10) — that transferring measured evidence is often more defensible than reconstructing it from an uncalibrated inversion.

8.2 Future Work

Priority items are executing the hindcasting protocol (Section 3.13.3) once further years extend the dataset past its current single-cutoff limitation; extending ingestion beyond 2025 so recently-matched twins accumulate the future years needed to resolve *uncertain* cells; adding independent field-ecological validation for the Ramsar sites; and migrating trajectory search to pgvector’s native ANN operators, since the current brute-force scan will not remain interactive as the cohort grows. Further directions

include calibrating the cosine-similarity display scale against labelled ecosystem pairs, attention- or learned-weighted pooling in place of coverage-weighted mean, growing the active cohort beyond 829 lakes, reducing monsoon-driven cloud gaps in temporal compositing, and deploying shared production infrastructure for multi-user access.

8.3 Final Remarks

EcoTwin’s central idea is to use historically observed ecosystems as computationally retrievable analogs rather than treating every ecosystem as an isolated forecasting problem. The Ramsar case studies show that this idea, carried through a fully processed 829-lake, ten-year cohort, produces results that are specific, checkable against independent knowledge of the sites where such knowledge exists, and — where the evidence ran out — honest about the limits of what it can currently claim.

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